

Materials Science 1 – Imperfections in Solids

Part 1

Session objectives

- Describe vacancies and self-interstitial crystalline defects;
- Identify relationships between vacancies and energy in crystal structures;
- Study defects in metals including solid solutions.

Learning outcomes

- Estimate the number of vacancy crystalline defects using first principles;
- Calculate the equilibrium number of vacancies and energy of formation in a material at some specified temperature, given the relevant constants.
- Predict the formation of a solid solution for metal materials.

Poll – Answers in WeChat

Q1) A crystalline defect can refer to a lattice irregularity that has one or more of its dimensions on the order of an atomic diameter

A) True

B) False

Poll – Answer

Q1) A crystalline defect can refer to a lattice irregularity that has one or more of its dimensions on the order of an atomic diameter

A) True

In a crystalline structure, a defect can be represented by an irregularity in the lattice for example due to the absence of one atom. Therefore, the dimension of the irregularity will be on the order of an atomic diameter.

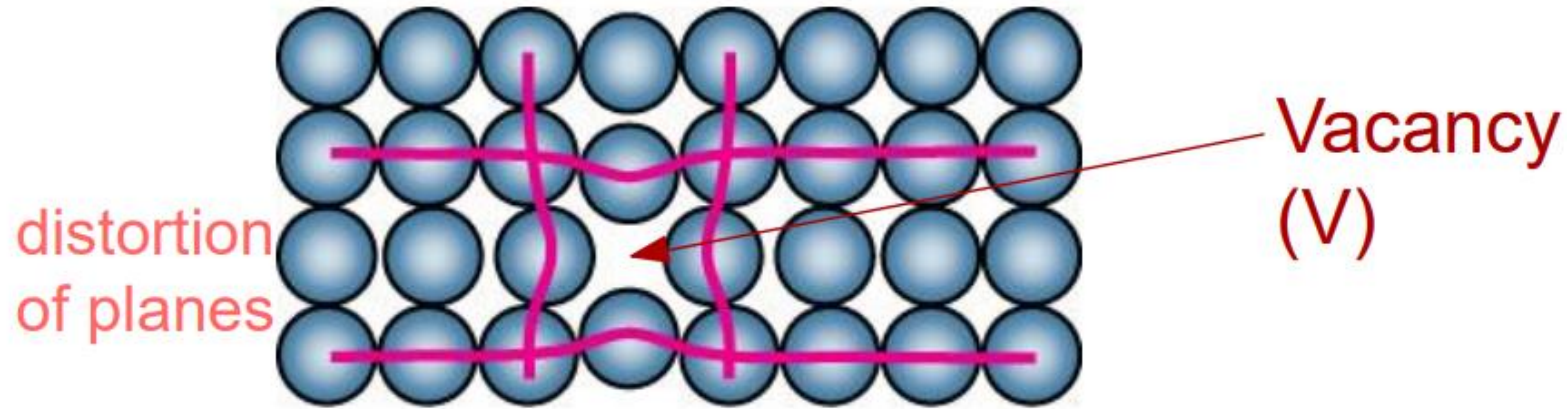
Types of imperfections in solids

- Point defects
 - Vacancy atoms
 - Interstitial atoms
 - Substitutional atoms
- Line defects
 - Dislocations
 - Antiphase boundaries
- Area Defects
 - Grain boundaries

Point defects in metals

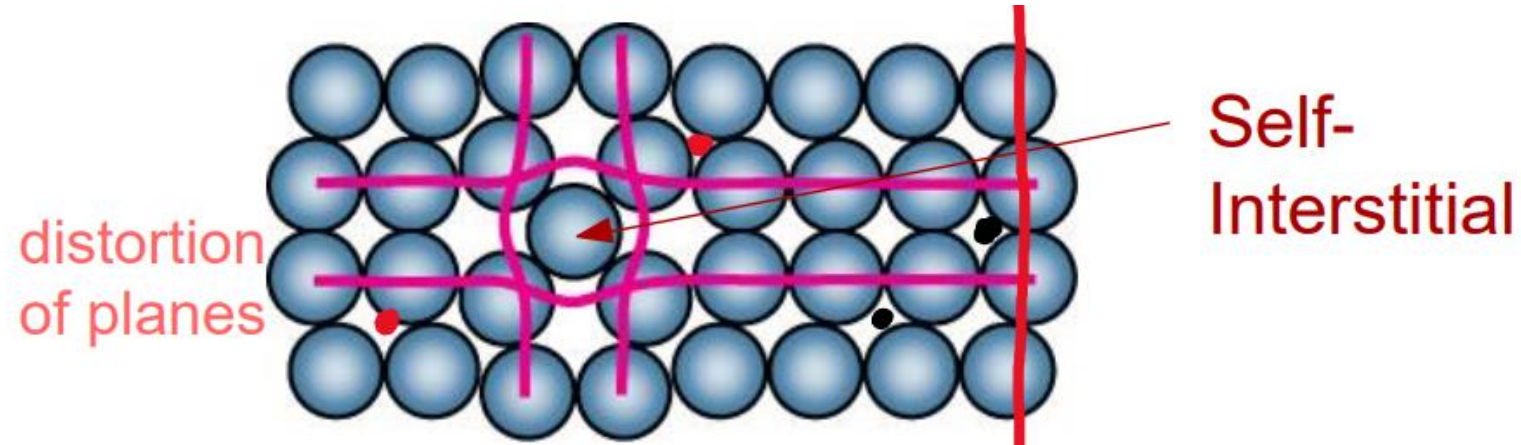
- Vacancies:

Vacant atomic sites in a structure



- Self-Interstitials:

"Extra" atoms positioned between atomic sites.



Vacancies

- Vacant lattice site that is normally occupied by an atom, which is missing;
- All crystalline solids contain vacancies; it is not possible to create a material without vacancies, because the presence of these defects increases the entropy (i.e. the disorder) of the crystal.
- The equilibrium number of vacancies (N_v) for a given quantity of material depends on and increases with temperature according to the following equation:

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$$

N = total number of atomic sites

Q_v = energy of formation of vacancy

T = absolute temperature (K)

k = Boltzmann's constant 1.38×10^{-23} J/atom-K

- The relationship above indicates that the number of vacancies increases exponentially with temperature. For most metals, the fraction of vacancies N_v/N just below the melting temperature is on the order of 10^{-4} : one lattice site out of 10,000 will be empty.

Vacancies

Example 1: Determining the number of vacancies N_v

The equilibrium fraction of lattice sites that are vacant in silver (Ag) at 700°C is 2×10^{-6} . Calculate the number of vacancies (per m^3) at 700°C.

1. Calculate N , number of atomic sites/ m^3 . Assume a density of 10.35 g/cm^3 for Ag.
2. Determine the number of atomic sites vacant taking into account the fraction of vacancies at 700°C.

Vacancies

Example 1 solution: Determining the number of vacancies N_v

The equilibrium fraction of lattice sites that are vacant in silver (Ag) at 700°C is 2×10^{-6} . Calculate the number of vacancies (per m^3) at 700°C.

1. Calculate N , number of atomic sites/ m^3 . Assume a density of 10.35 g/cm^3 for Ag.
2. Determine the number of atomic sites vacant taking into account the fraction of vacancies at 700°C.

$$\rho = \frac{nA}{V_C N_A}$$

1. Total number of lattice sites in Ag:

$$N_{\text{Ag}} = \frac{N_A \rho_{\text{Ag}}}{A_{\text{Ag}}} = \frac{(6.022 \times 10^{23} \text{ atoms/mol})(10.35 \text{ g/cm}^3)(10^6 \text{ cm}^3/\text{m}^3)}{107.87 \text{ g/mol}}$$

2. Number of vacancies/ m^3 in Ag:

$$N_v = (2 \times 10^{-6}) N_{\text{Ag}} = (2 \times 10^{-6})(5.78 \times 10^{28} \text{ atoms/m}^3) = 1.156 \times 10^{23} \text{ vacancies/m}^3$$

Vacancies

Example 2: Determining the number of vacancies N_v

For a hypothetical metal, the equilibrium number of vacancies at 900°C is $2.3 \times 10^{25} \text{ m}^{-3}$. If the density and atomic weight of this metal are 7.40 g/cm³ and 85.5 g/mol, respectively, calculate the fraction of vacancies for this metal at 900°C.

Vacancies

Example 2 solution: Determining the number of vacancies N_v

For a hypothetical metal, the equilibrium number of vacancies at 900°C is $2.3 \times 10^{25} \text{ m}^{-3}$. If the density and atomic weight of this metal are 7.40 g/cm^3 and 85.5 g/mol , respectively, calculate the fraction of vacancies for this metal at 900°C.

1. Calculate the total number of sites

$$N = \frac{N_A \rho}{A} = \frac{(6.022 \times 10^{23} \text{ atoms/mol})(7.40 \text{ g/cm}^3)(10^6 \text{ cm}^3/\text{m}^3)}{85.5 \text{ g/mol}}$$
$$= 5.21 \times 10^{28} \text{ atoms/m}^3$$

2. Calculate the fraction of vacancies

$$\frac{N_v}{N} = \frac{2.3 \times 10^{25} \text{ m}^{-3}}{5.21 \times 10^{28} \text{ atoms/m}^3} = 4.41 \times 10^{-4}$$

Vacancies

Equilibrium concentration of vacancies N_v varies with temperature:

No. of vacancies $\rightarrow \frac{N_v}{N} = \exp \left(\frac{-Q_v}{kT} \right)$

No. of potential vacancies sites $\rightarrow N$

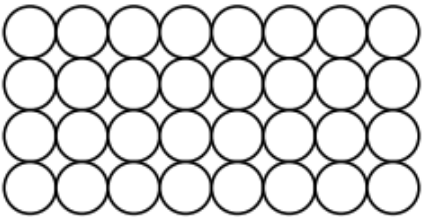
Activation energy $\rightarrow Q_v$

Boltzmann's constant $\rightarrow k$

Temperature $\rightarrow T$

$(1.38 \times 10^{-23} \text{ J/atom-K})$
 $(8.62 \times 10^{-5} \text{ eV/atom-K})$

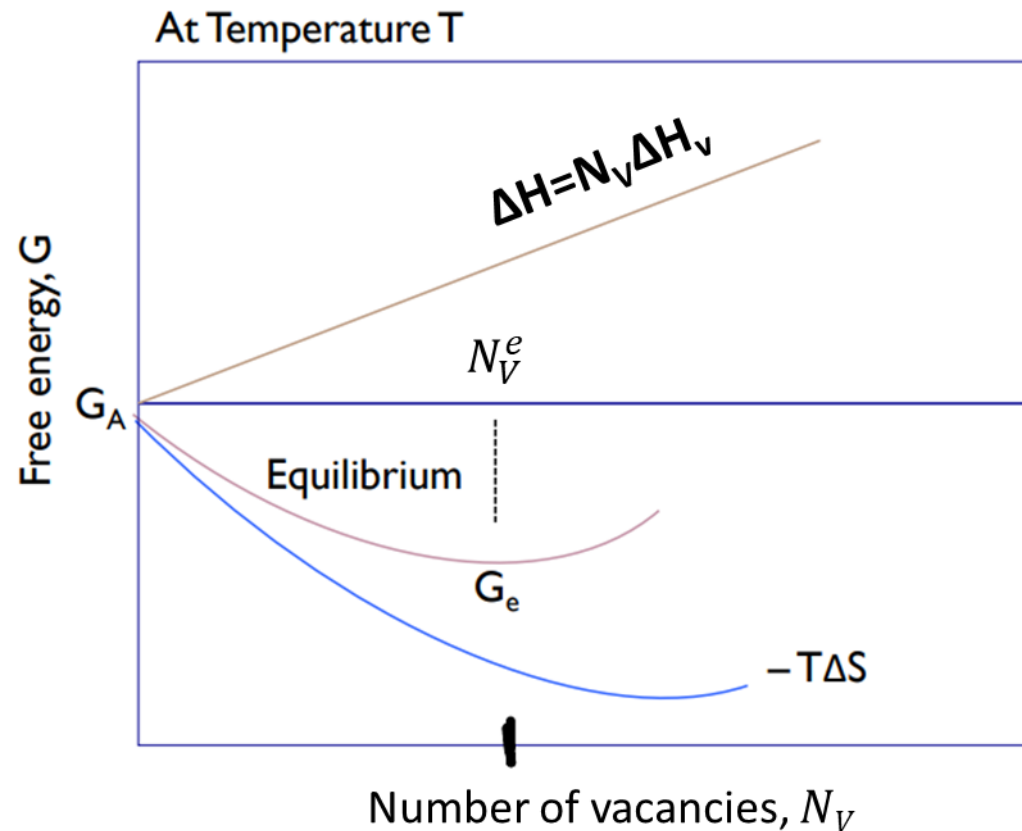
Each lattice site is a potential vacancy site $\rightarrow$ So N can be regarded as the total number of lattice sites as seen earlier.



Vacancies

Why we say “equilibrium concentration” of vacancies?

- In thermodynamics, equilibrium conditions are identified as states of minimum Gibbs free energy (G) at a given temperature and pressure.
- A crystal with vacancies has lower Gibbs free energy than a perfect crystal.



The Gibbs free energy G is defined as:

$$G = H - TS$$

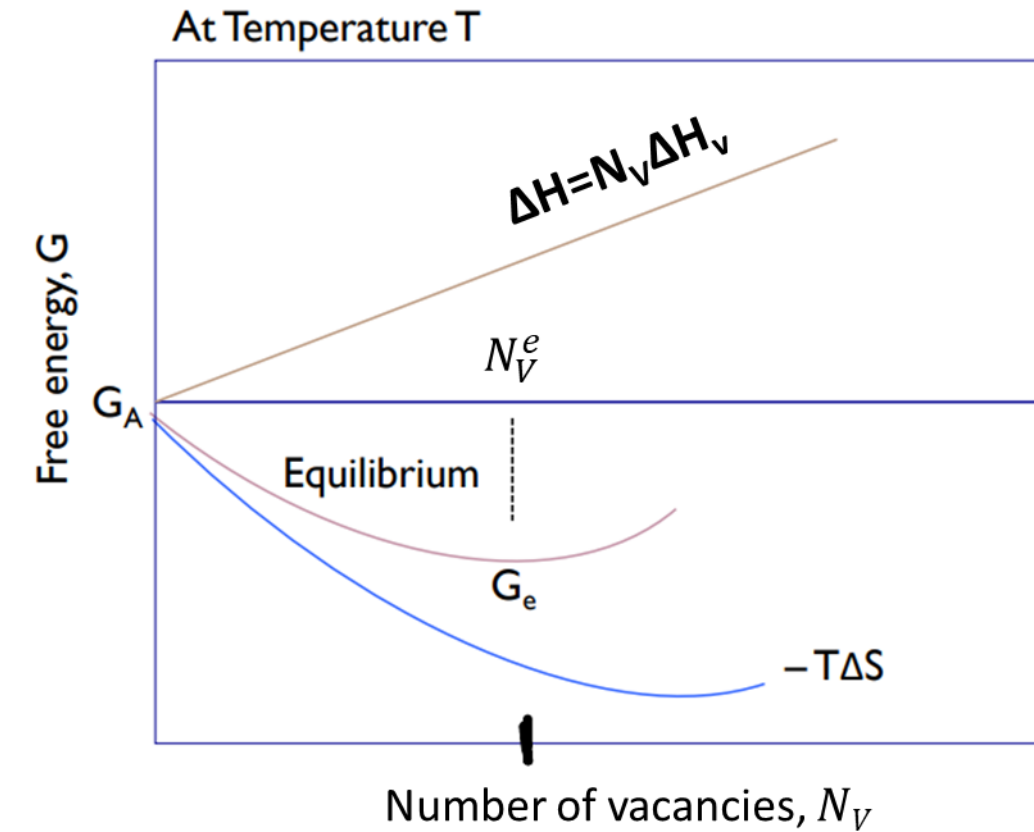
Where H is the enthalpy, S the entropy and T the absolute temperature.

Starting from a state A with free energy G_A , the free energy G in a state after vacancies formation can be written as:

$$G = G_A + \Delta G = G_A + \Delta H - T\Delta S$$

Vacancies

Why we say “equilibrium concentration” of vacancies?



$$G = G_A + \Delta G = G_A + \Delta H - T\Delta S$$

$$\Delta H = N_v \Delta H_v$$

The formation of vacancies increases the enthalpy of the crystal. ΔH_v indicate the enthalpy change due to the formation of only 1 vacancy.

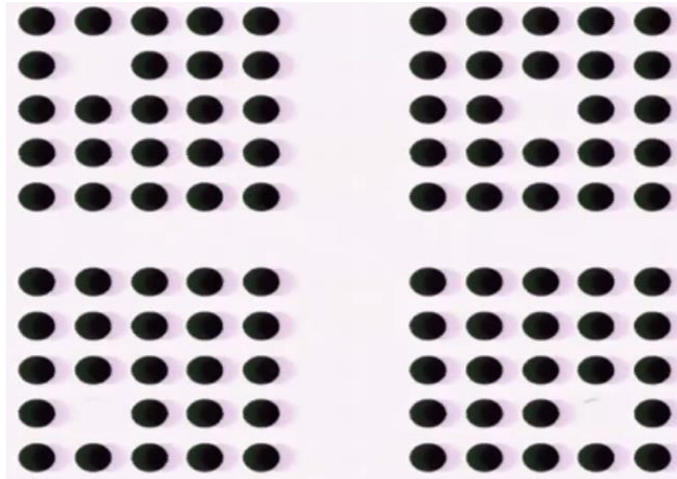
$$\Delta S = k \ln W$$

k is the Boltzmann's constant and W is the number of microstates corresponding to a given macrostate.

Vacancies

Why we say “equilibrium concentration” of vacancies?

$$\Delta S = k \ln W$$



Example of 4 different locations for a vacancy (microstate) in a regular group of 25 atoms.

Macroscopically, the different position of the vacancy is not expected to create changes in the thermodynamic state of the system (one given macrostate).

For a system with N atoms and N_V vacancies, the total number of microstates W (different microscopic configuration) is given by:

$$W = \frac{(N + N_V)!}{N! N_V!}$$

Substituting the expression of W in ΔS :

$$\Delta S = k \ln W = k \ln \frac{(N + N_V)!}{N! N_V!} = k [\ln(N + N_V)! - \ln(N! N_V!)]$$

According to the Stirling approximation:

$$\ln N! \approx N \ln N - N$$

We have:

$$\Delta S = k [(N + N_V) \ln(N + N_V) - N_V \ln N_V - N \ln N]$$

Vacancies

Why we say “equilibrium concentration” of vacancies?

Substituting the expressions of ΔH and ΔS in G :

$$G = G_A + \Delta G = G_A + \Delta H - T\Delta S$$

we have:

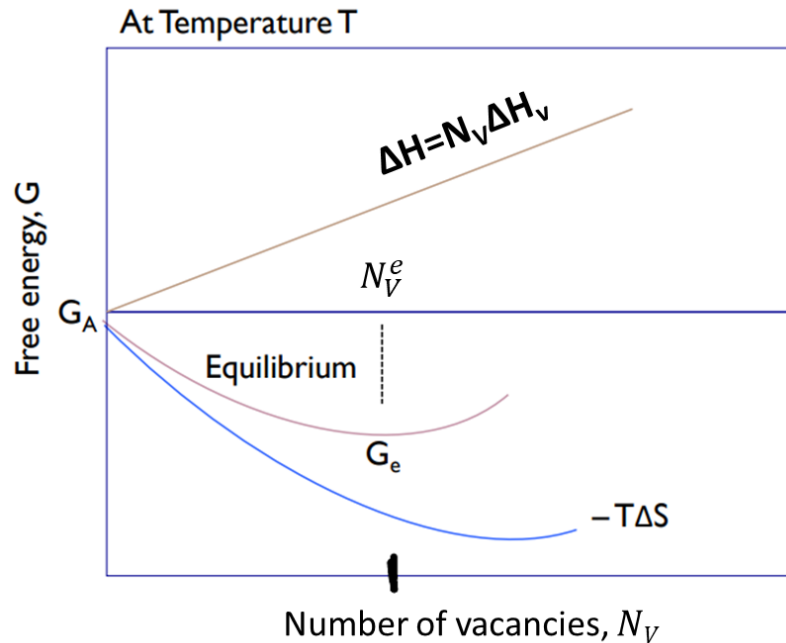
$$G = G_A + N_V \Delta H_V - Tk[N \ln(N + N_V) + N_V \ln(N + N_V) - N_V \ln N_V - N \ln N]$$

To find the equilibrium state, we need to find the minimum of G : $\frac{\partial G}{\partial N_V} = 0$

By skipping some passages of the derivative and some algebra we have:

$$\Delta H_V - Tk \left[\ln \left(\frac{N + N_V}{N_V} \right) \right] = 0$$

$$\frac{N_V}{N + N_V} = N_V^e = \exp \left(- \frac{\Delta H_V}{kT} \right)$$



Vacancies

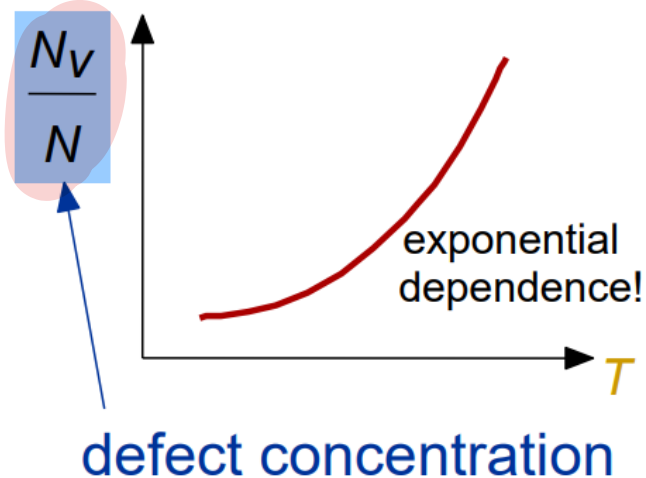
Measurements

Combinations of appropriate experimental tests allow characterize the formation of vacancies as a function of temperature. Typical experiments are represented by dilatometry (thermal expansion coefficient) as a function of temperature and X-ray diffraction (XRD):

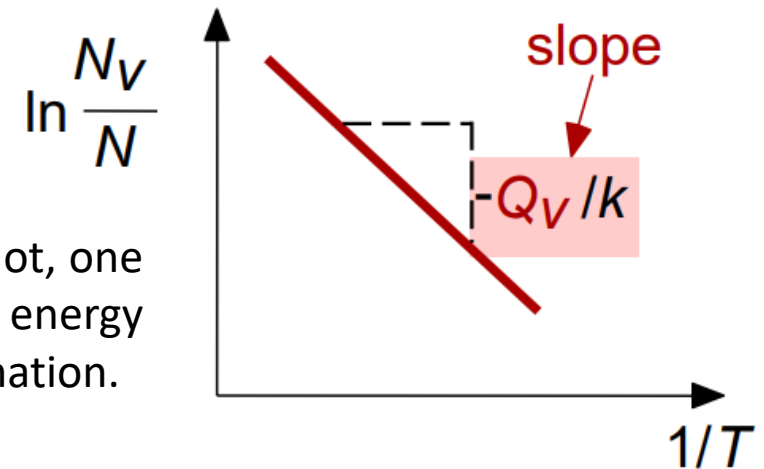
$$\frac{\Delta V}{V} = \frac{3\Delta L}{L} = \frac{\Delta v}{v} + \frac{N_V}{N}$$

- Macroscopic expansion (in volume V or length L) is due to expansion of the unit cell volume v and the variation of vacancies concentration X_V .
- Macroscopic expansion can be measured by dilatometry, while unit cells volume change can be obtained by XRD at different temperatures.

$$\frac{N_V}{N} = \exp\left(-\frac{Q_V}{kT}\right)$$



From the logarithmic plot, one can get the activation energy Q_V for the vacancy formation.



Vacancies

Example 3: Estimating Vacancy Concentration

Determine the number of vacancies at equilibrium in 1 m³ of Cu at 1000°C.

Use the data below for your calculations.

$$\rho = 8.4 \text{ g/cm}^3;$$

$$A_{\text{Cu}} = 63.5 \text{ g/mol};$$

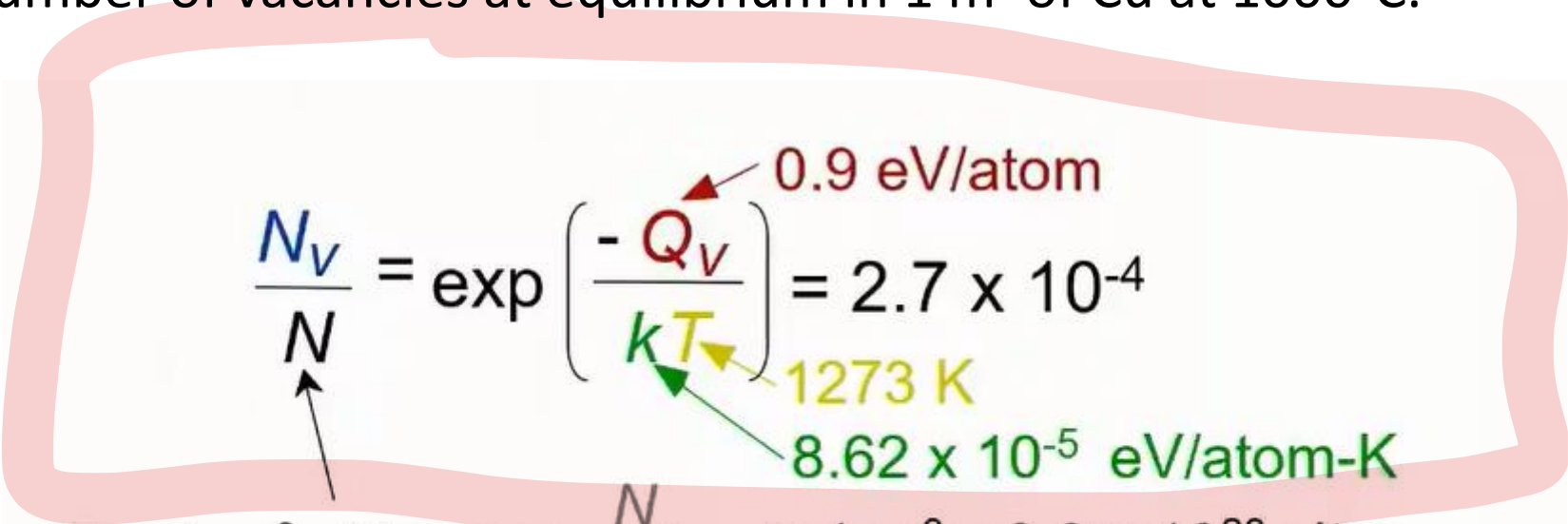
$$Q_v = 0.9 \text{ eV/atom};$$

$$N_A = 6.02 \times 10^{23} \text{ atoms/mol}$$

Vacancies

Example 3 solution: Estimating Vacancy Concentration - Solution

Determine the number of vacancies at equilibrium in 1 m³ of Cu at 1000°C.



The diagram shows the equation $\frac{N_V}{N} = \exp\left(\frac{-Q_V}{kT}\right) = 2.7 \times 10^{-4}$ enclosed in a red hand-drawn box. Annotations include: a red arrow pointing to Q_V with the text "0.9 eV/atom"; a yellow arrow pointing to T with the text "1273 K"; and a green arrow pointing to k with the text "8.62 x 10⁻⁵ eV/atom-K".

$$\frac{N_V}{N} = \exp\left(\frac{-Q_V}{kT}\right) = 2.7 \times 10^{-4}$$

For 1 m³, $N = \rho \times \frac{N_A}{A_{\text{Cu}}} \times 1 \text{ m}^3 = 8.0 \times 10^{28}$ sites

$$N_V = (2.7 \times 10^{-4})(8.0 \times 10^{28}) \text{ sites} = 2.2 \times 10^{25} \text{ vacancies}$$

Vacancies

Example 4: Estimating energy for vacancy formation

Calculate the energy for vacancy formation in nickel (Ni), given that the equilibrium number of vacancies at 850°C (1123 K) is $4.7 \times 10^{22} \text{ m}^{-3}$. The atomic weight and density (at 850°C) for Ni are, respectively, 58.69 g/mol and 8.80 g/cm³.

Vacancies

Example 4: Estimating energy for vacancy formation - Solution

Calculate the energy for vacancy formation in nickel (Ni), given that the equilibrium number of vacancies at 850°C (1123 K) is $4.7 \times 10^{22} \text{ m}^{-3}$. The atomic weight and density (at 850°C) for Ni are, respectively, 58.69 g/mol and 8.80 g/cm³.

1. Calculating N for Ni:

$$N = \frac{N_A \rho_{\text{Ni}}}{A_{\text{Ni}}} = \frac{(6.022 \times 10^{23} \text{ atoms/mol})(8.80 \text{ g/cm}^3)}{58.69 \text{ g/mol}}$$

$$N = 9.03 \times 10^{22} \text{ atoms/cm}^3 = 9.03 \times 10^{28} \text{ atoms/m}^3$$

2. Using the relationship for N_v/N and solving for Q_v (Energy):

$$\frac{N_v}{N} = \exp \frac{-Q_v}{kT}$$

$$\ln N_v = \ln N - \frac{Q_v}{kT} \quad \longrightarrow \quad Q_v = -kT \ln \left| \frac{N_v}{N} \right|$$

Vacancies

Example 4: Estimating energy for vacancy formation - Solution

Calculate the energy for vacancy formation in nickel (Ni), given that the equilibrium number of vacancies at 850°C (1123 K) is $4.7 \times 10^{22} \text{ m}^{-3}$. The atomic weight and density (at 850°C) for Ni are, respectively, 58.69 g/mol and 8.80 g/cm³.

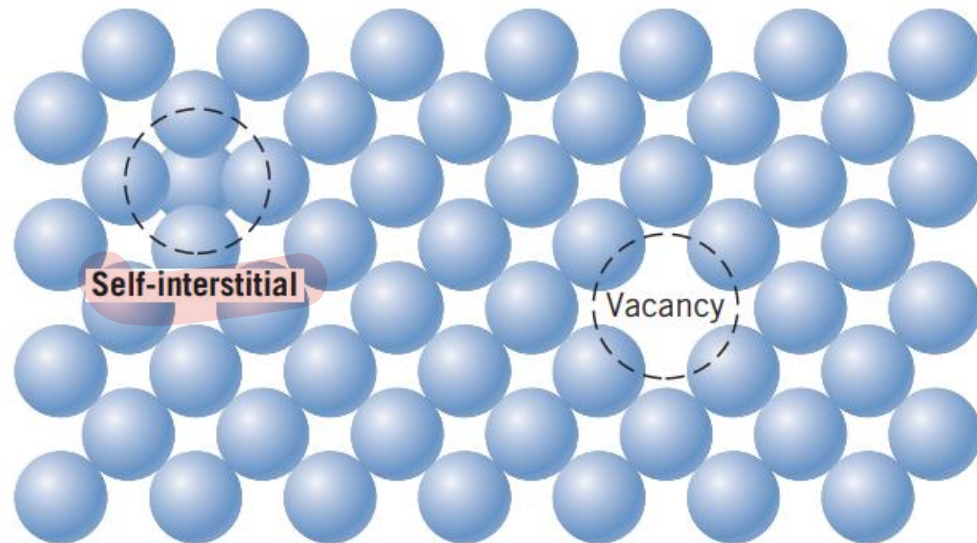
3. Incorporating all data into Q_v equation, using all the data provided:

$$Q_v = - (8.62 \times 10^{-5} \text{ eV/atom-K})(1123 \text{ K}) \ln \left[\frac{4.7 \times 10^{22} \text{ m}^{-3}}{9.03 \times 10^{28} \text{ m}^{-3}} \right]$$

$$Q_v = 1.40 \text{ eV/atom}$$

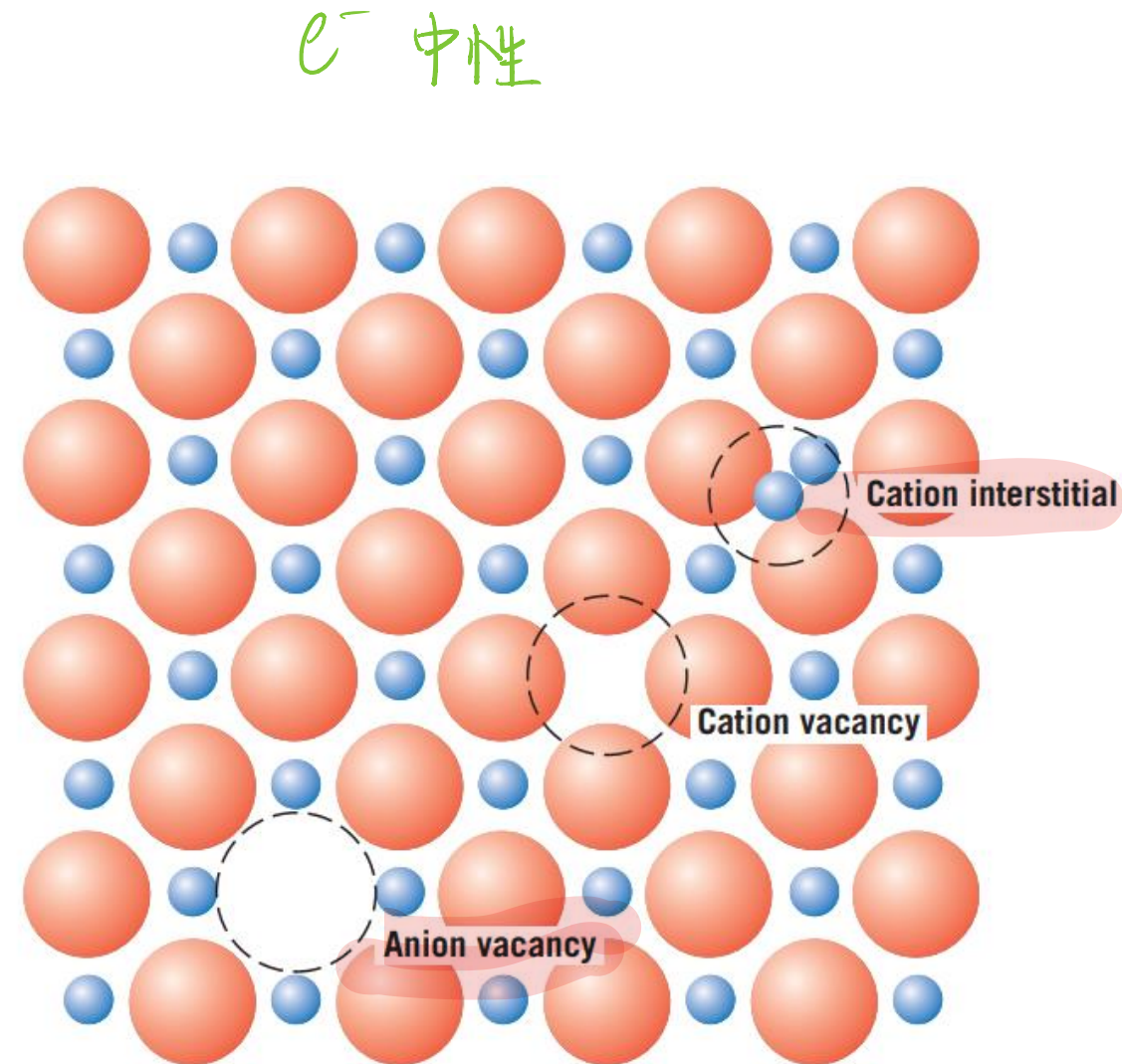
Self-interstitials

- A **self-interstitial** is an atom from the crystal that is located into an interstitial site, which would not be occupied in normal circumstances.
- In metals, a self-interstitial introduces relatively large distortions in the surrounding lattice because the atom is substantially larger than the interstitial position in which it is situated.
- Consequently, the formation of this defect is not highly probable, and it exists in very small concentrations that are significantly lower than for vacancies.



Point defects in ceramics

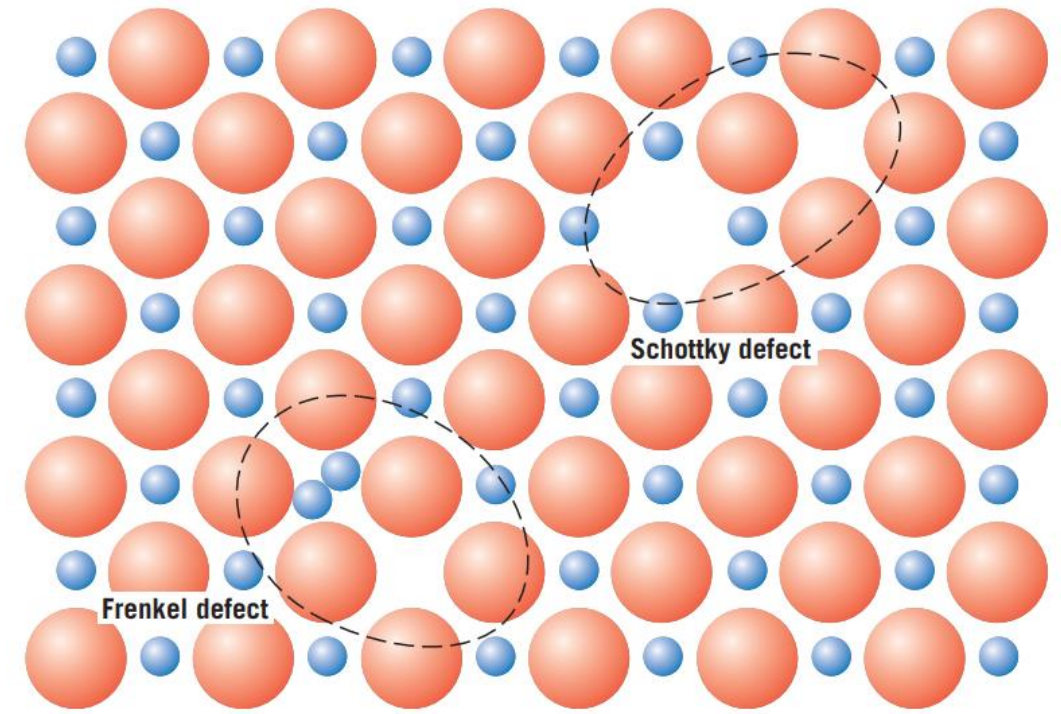
- As with metals, both **vacancies and interstitials are possible** for each ion type.
- However, it is rather improbable that there would be appreciable concentrations of anion interstitials as the anion is relatively large.
- The expression defect structure is often used to designate the types and concentrations of atomic defects in ceramics.
- Because the atoms exist as charged ions, when defect structures are considered, conditions of **electroneutrality must be maintained**.



Point defects in ceramics

Schottky and Frenkel defects

- Due to electroneutrality condition, defects in ceramics do not occur alone.
- A type of these defect structures is represented by a cation vacancy and a cation interstitial pair, known as **Frenkel defect**.
- Another type of defect structures found in ceramics is constituted by a cation vacancy and anion vacancy pair, known as **Schottky defect**.
- The ratio of cations to anions is not altered by the formation of a Frenkel or a Schottky defect. If no other defects are present, the material is said to be **stoichiometric**.

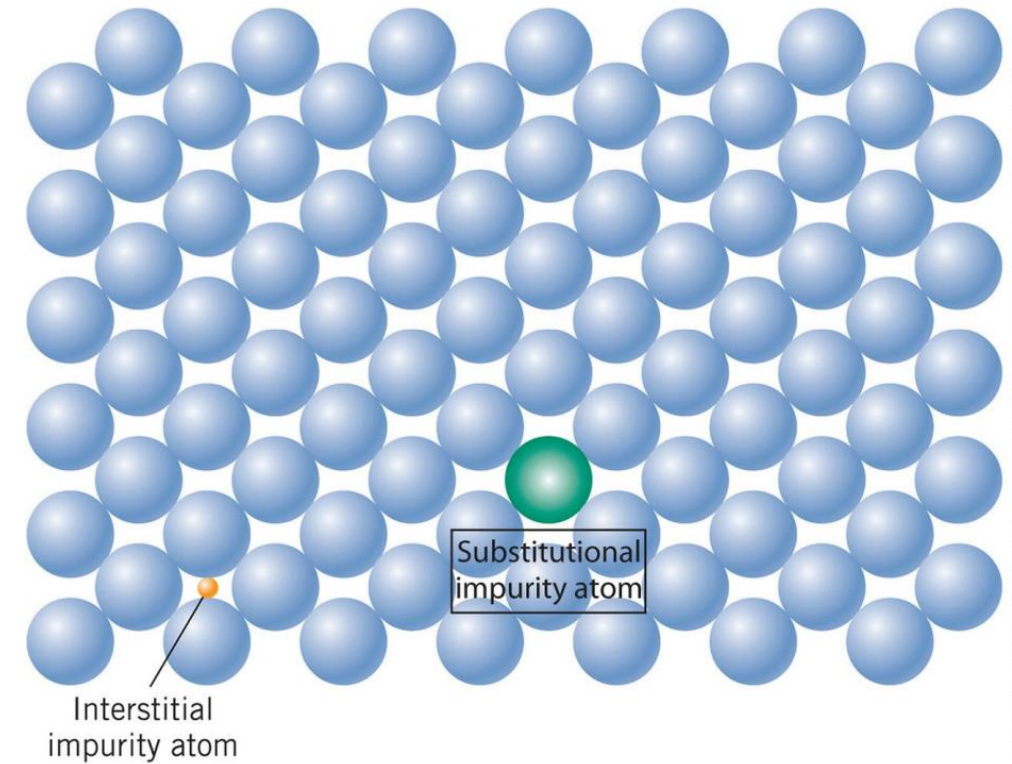


Frenkel defects: found in compounds with low coordination number and large size difference of cations and anions.

Schottky defects: found in highly ionic compounds with high coordination number and small size difference of cations and anions.

Impurities in solids

- A **pure metal** consisting of only one type of atom does not exist, it would always contain other elements and some exist as crystalline point defects (i.e. interstitials).
- Most metals are rather **alloys**, as impurity atoms have been added intentionally to impart specific characteristics to the material (i.e. improve mechanical strength and corrosion resistance).
- The addition of impurity atoms (**alloying elements or solute atoms**) to a metal results in the formation of a **solid solution** and/or **new phase(s)**, often named **secondary phases**, depending on the kinds of impurity, their concentrations, and the temperature of the alloy.



Adapted from W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, Structure, p. 77. Copyright © 1964 by John Wiley & Sons, New York, NY. Reprinted by permission of John Wiley & Sons, Inc.

超过溶极限 不能继续溶
出现 secondary phase

Materials Science 1 – Imperfections in Solids

Part 2

Session objectives

Describe linear and interfacial defects:

- Edge and screw dislocations;
- Mixed dislocations;
- Grain boundaries;
- Phase boundaries;
- Twin boundaries.

Learning outcomes

- Describe and make a drawing of the dislocation;
- Describe and make drawings of interfacial defects;
- Understand the importance of defects in the applications of materials.

Poll – Answers in WeChat

Q1 - What is a point defect?

- a) A defect that occurs in crystal structures and is associated with one or two atomic positions.
- b) A defect that occurs in crystal structures only in one point across the whole structure.

Poll – Answer

Q1 - What is a point defect?

- a) A defect that occurs in crystal structures and are associated with one or two atomic positions.

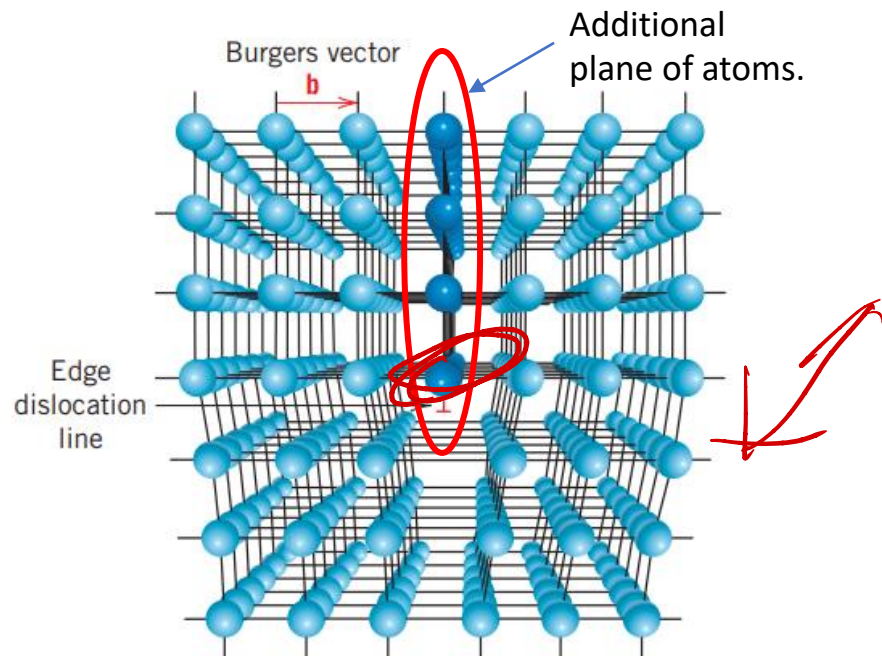
A point defect could occur in the form of a vacancy, a self-interstitial and it may occur in several points across the crystal structure.

Other types of imperfections in solids

Linear defects (or line defects)

Dislocations: most important line defects. These can be regarded as one-dimensional defects, around which atoms are misaligned. There are two types of dislocations:

- **Edge Dislocations:** an extra portion of a plane of atoms (**also called dislocation line**), or half-plane, the edge of which terminates within the **crystal**.



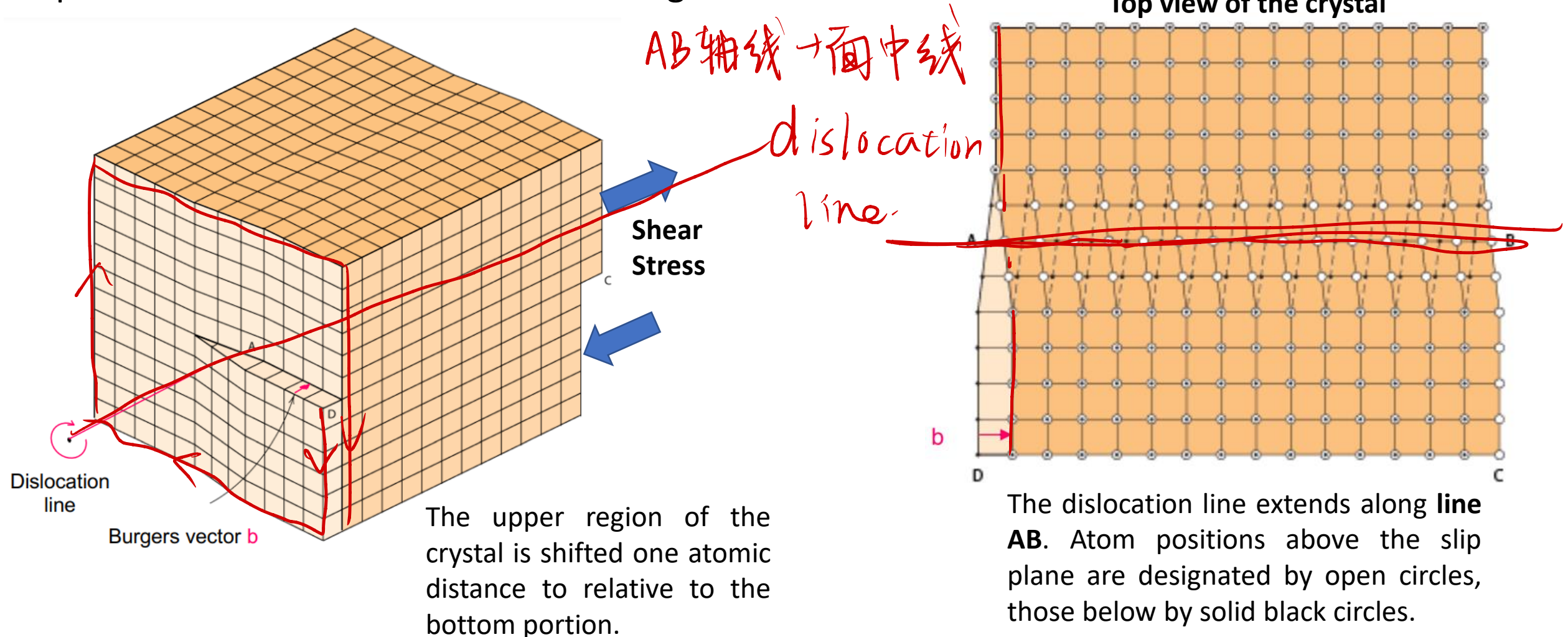
Observations:

- In the region around the dislocation line, there is some localized lattice distortion: the atoms in the lattice part containing the dislocation are squeezed, while those below are pulled apart.
- The magnitude of the distortion decreases moving away from the dislocation.
- Edge dislocations are usually represented by the symbol $\perp$, which also indicates the position of the dislocation line.

Other types of imperfections in solids

Linear defects (or line defects)

- Screw dislocations:** may be thought of as being formed by a shear stress that is applied to produce the distortion shown in the figures below.

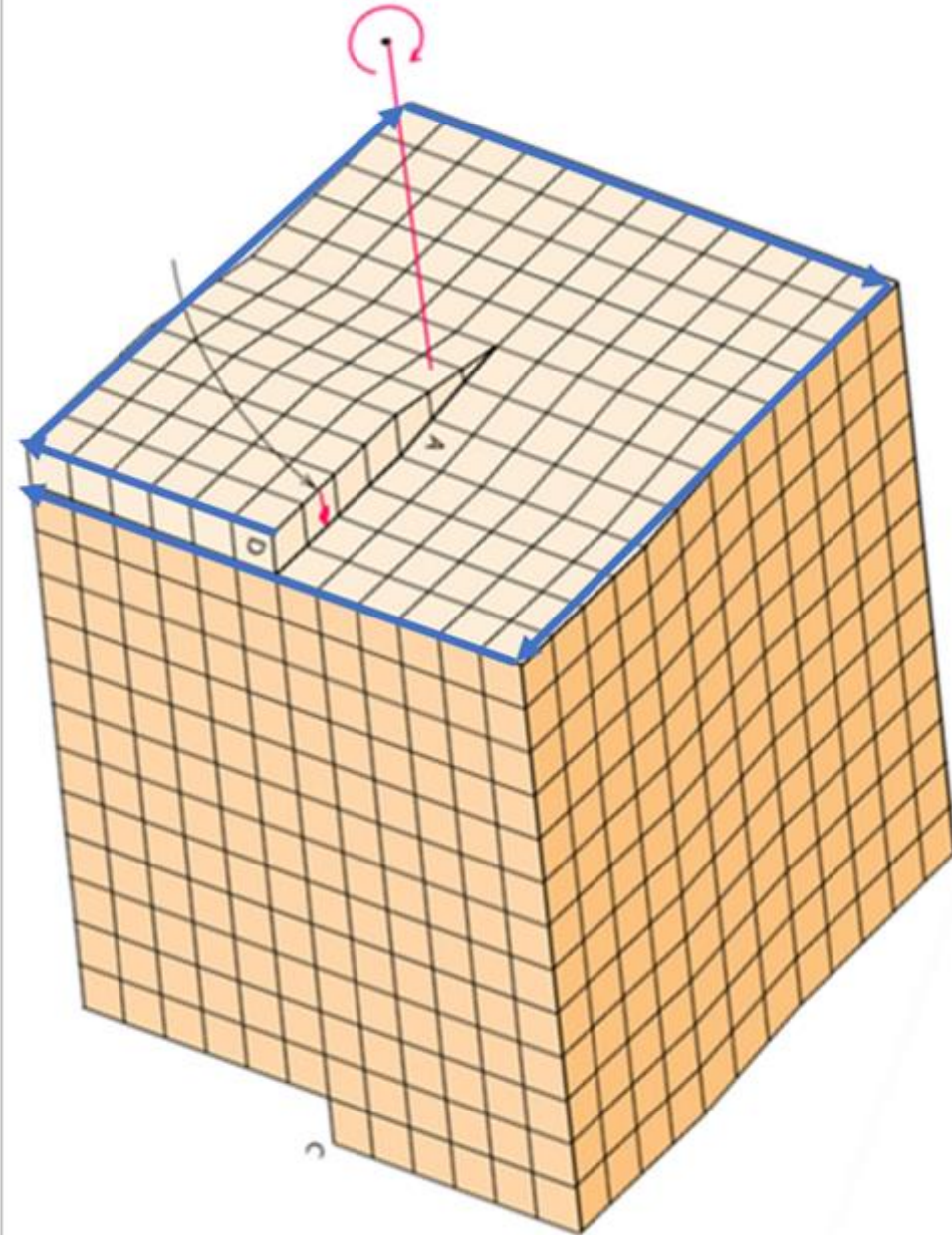


Other types of imperfections in solids

Linear defects (or line defects)

- **Why “screw dislocations”?**

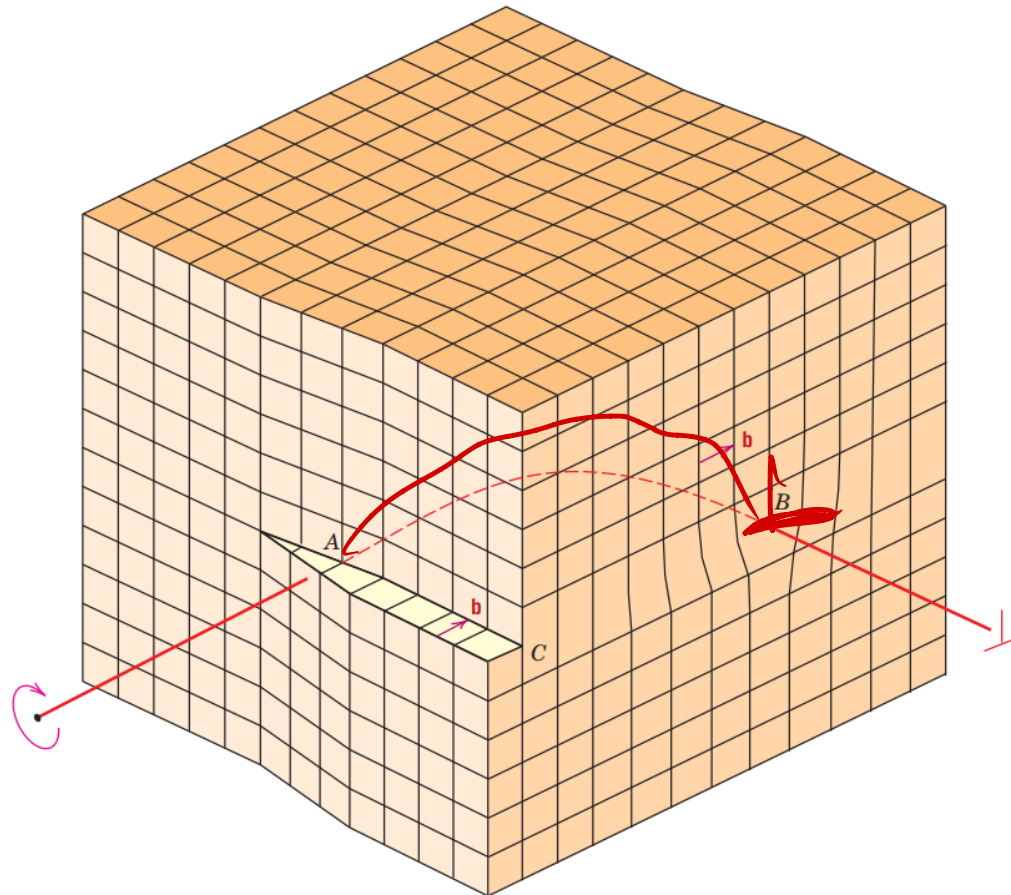
In a screw dislocation, the atomic positions follow a helical path where the dislocation line is the axis. Moving along atomic positions around the dislocation line it is similar to follow the threads in a screw.



Other types of imperfections in solids

Linear defects (or line defects)

- **Mixed dislocations:** dislocations that have mixed edge and screw character. At point A, the dislocation is pure screw, while at point B, it is pure edge. For regions in between where there is curvature in the dislocation line, the character is mixed edge and screw.



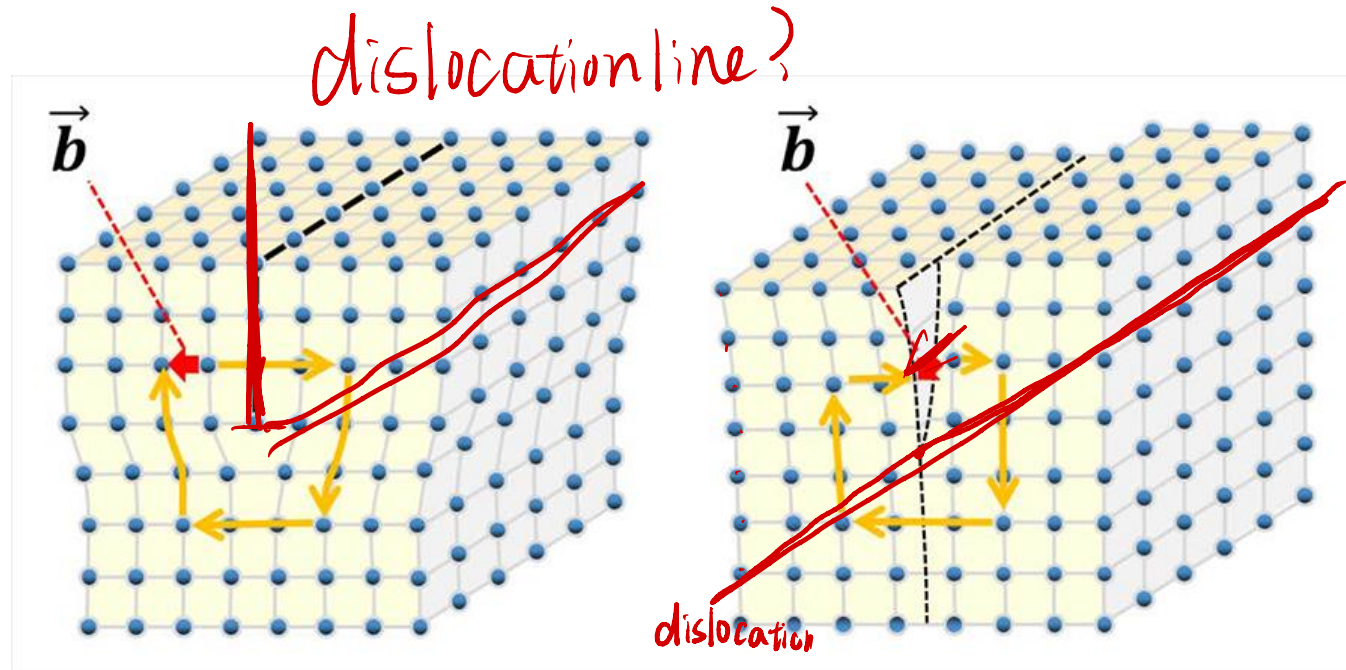
Other types of imperfections in solids

Linear defects (or line defects)

Burgers vector: expresses the magnitude and direction of the lattice distortion associated with a dislocation, usually denoted by $\vec{b}$. It can be also used to identify the type of dislocation (i.e., edge, screw, or mixed), which can be defined by the relative orientations of dislocation line and Burgers vector.

Edge dislocation:

$\vec{b}$ is perpendicular to the dislocation line.



Screw dislocation:

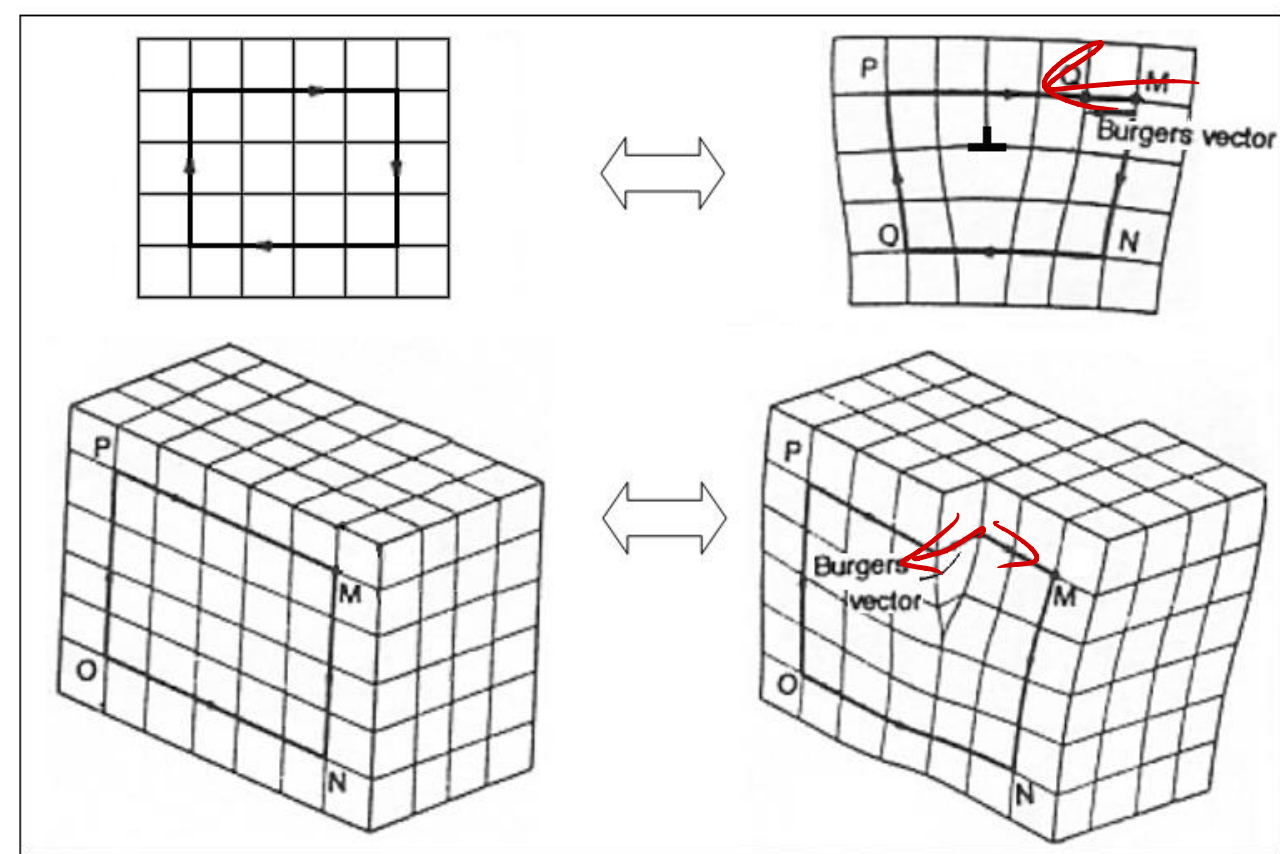
$\vec{b}$ is parallel to the dislocation line.

- For metallic materials, the Burgers vector for a dislocation points in a close-packed crystallographic direction and is of magnitude equal to the interatomic spacing.

Other types of imperfections in solids

Linear defects (or line defects)

Burgers circuit: method to identify magnitude and direction of the Burgers vector.



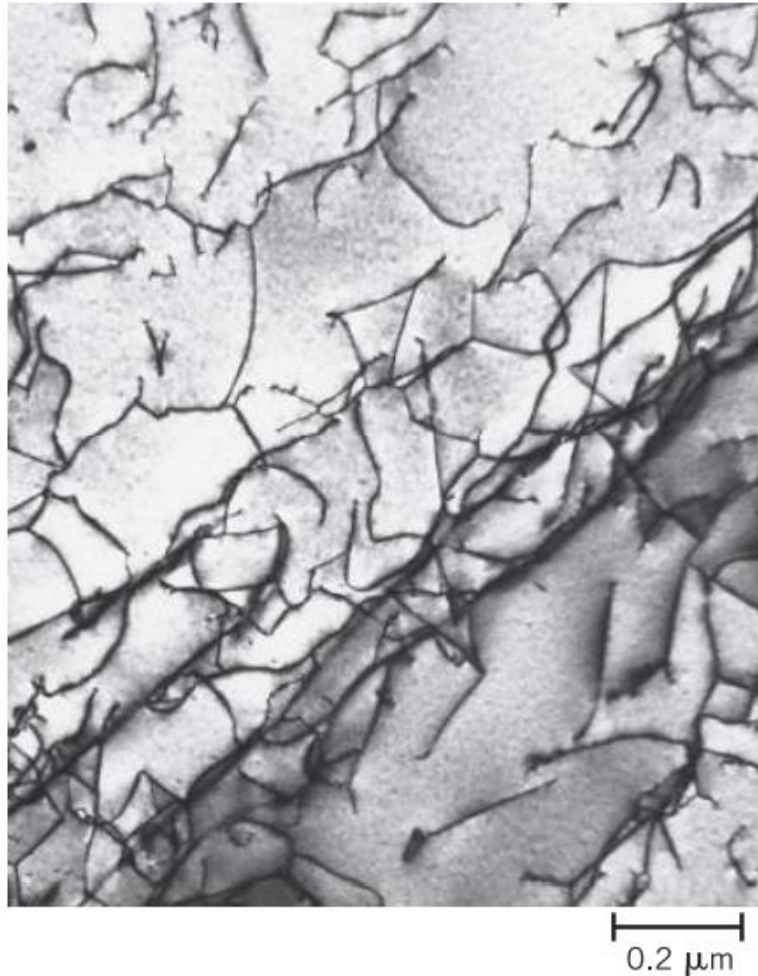
- In a perfect lattice, moving by the same number of atomic steps along horizontal and vertical directions, one returns to the same initial position.
- In presence of a dislocation, to return to the same initial position, an additional step is necessary. This is identified by the Burgers vector.
- In case of an edge dislocation, one can see that the Burgers vector is perpendicular to the dislocation line.
- In case of a screw dislocation the Burgers vector is perpendicular to the dislocation line.

Other types of imperfections in solids

Imaging dislocations

Slip → *plastic deformation*

Dislocations can be imaged by transmission electron microscopy.



Transmission electron micrograph of a titanium alloy in which the dark lines are dislocations.

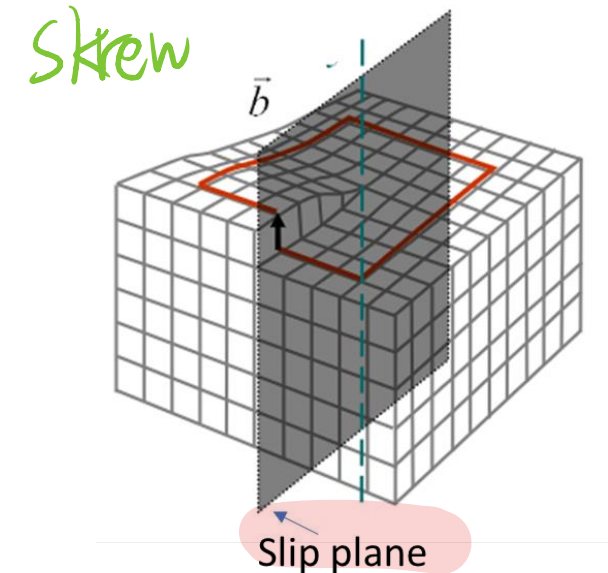
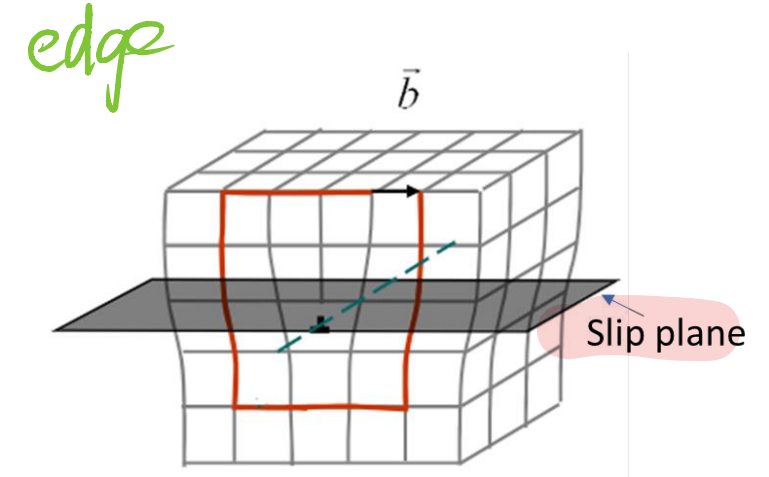
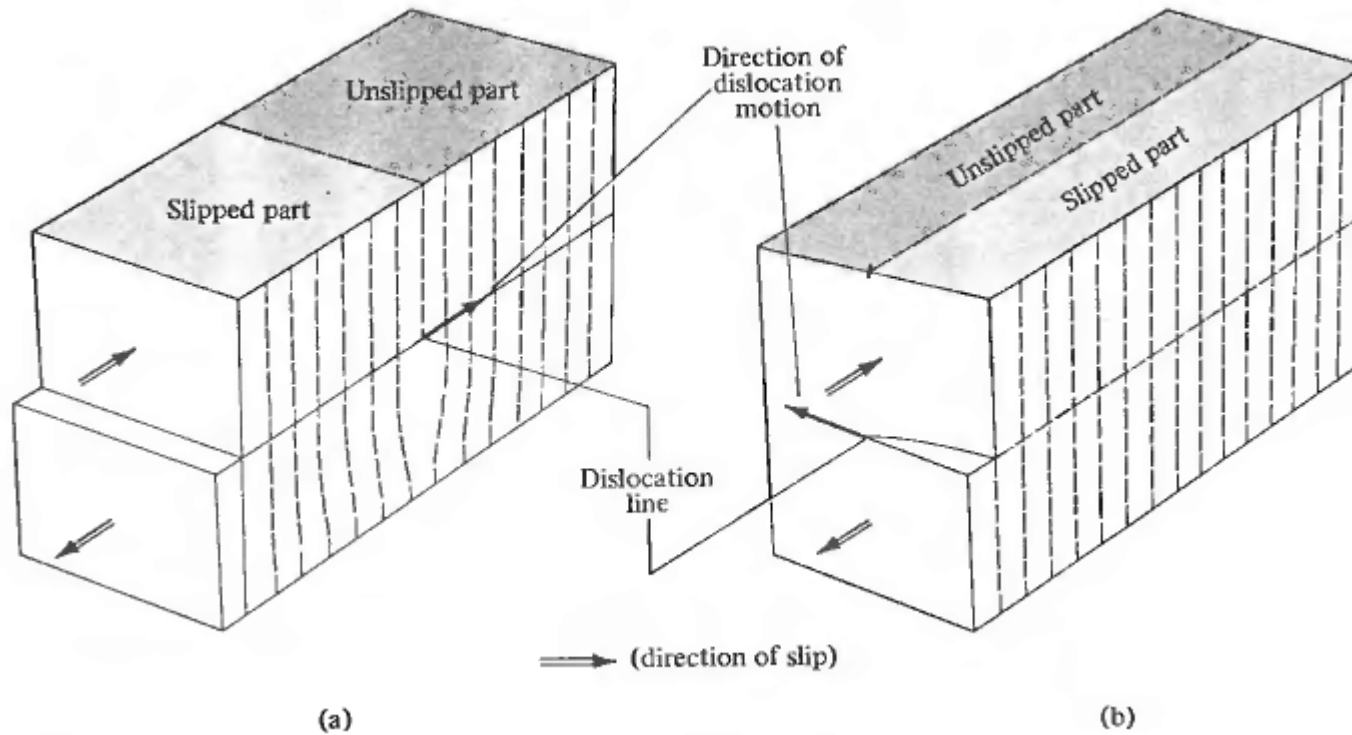
<https://www.youtube.com/watch?v=3VHKG GoyZQk>

Movement of dislocation during heating.

Other types of imperfections in solids

Movement of dislocations

Under applied stresses, dislocations can move on slip planes.



Other types of imperfections in solids

Movement of dislocations

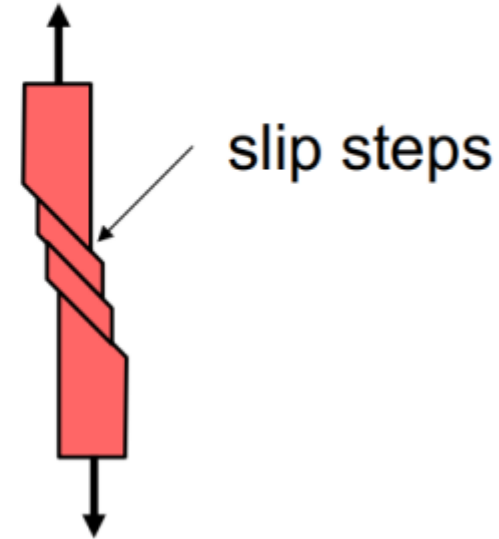
Macroscopically, the movement of dislocation on the slip planes produces plastic deformations (permanent or irreversible deformations).

Schematic of Zinc (HCP):

- before deformation



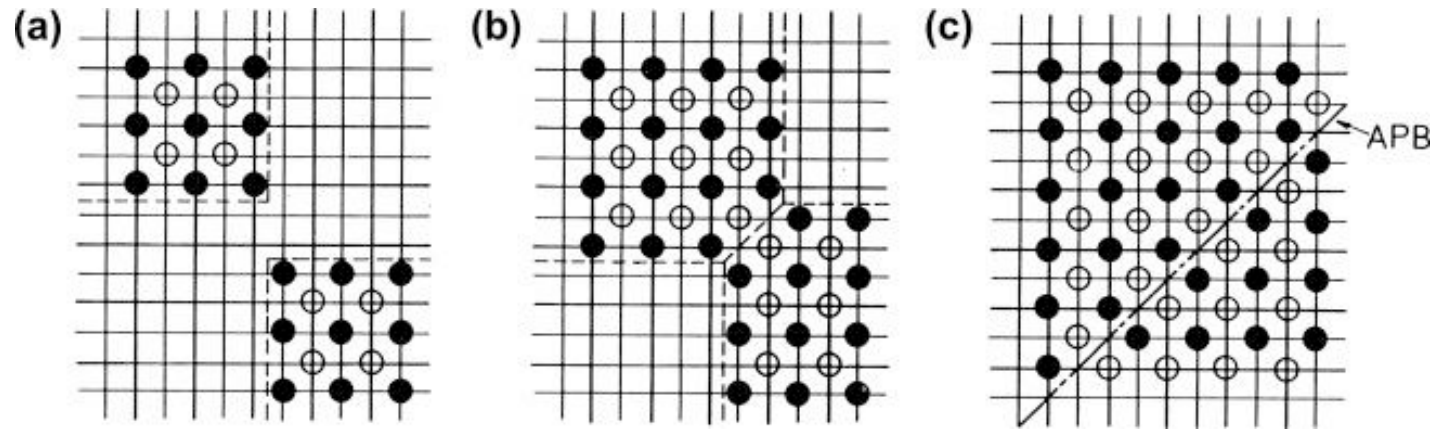
- after tensile elongation



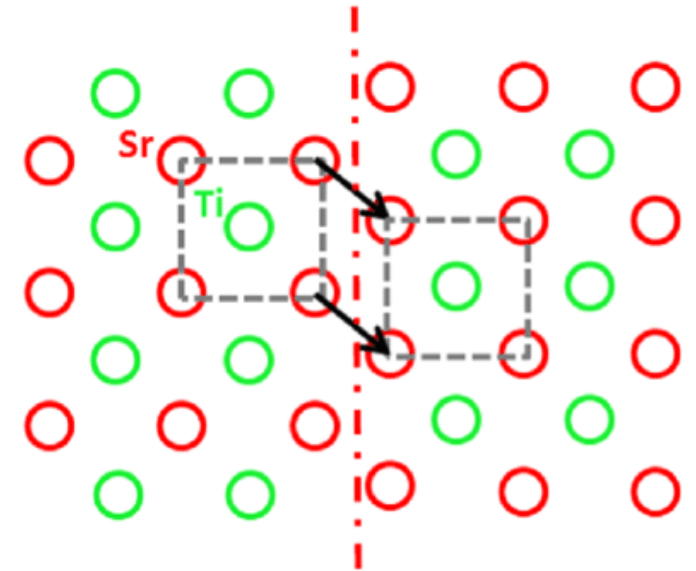
Other types of imperfections in solids

Antiphase boundaries

An antiphase boundary (APB) separates two domains of the same ordered phase.



Formation of an APB when ordered regions in which A (open circles) and B (filled circles) atoms occupying different sublattices grow together: (a) nucleation of ordered domains on different sublattices, (b) contact of domains, and (c) the resulting APB (dashed line).



Schematics of APB in SrTiO_3

Other types of imperfections in solids

Interfacial Defects

Interfacial defects are boundaries that have two dimensions and normally separate regions of the materials that have different crystal structures and/or crystallographic orientations. These types of defects include:

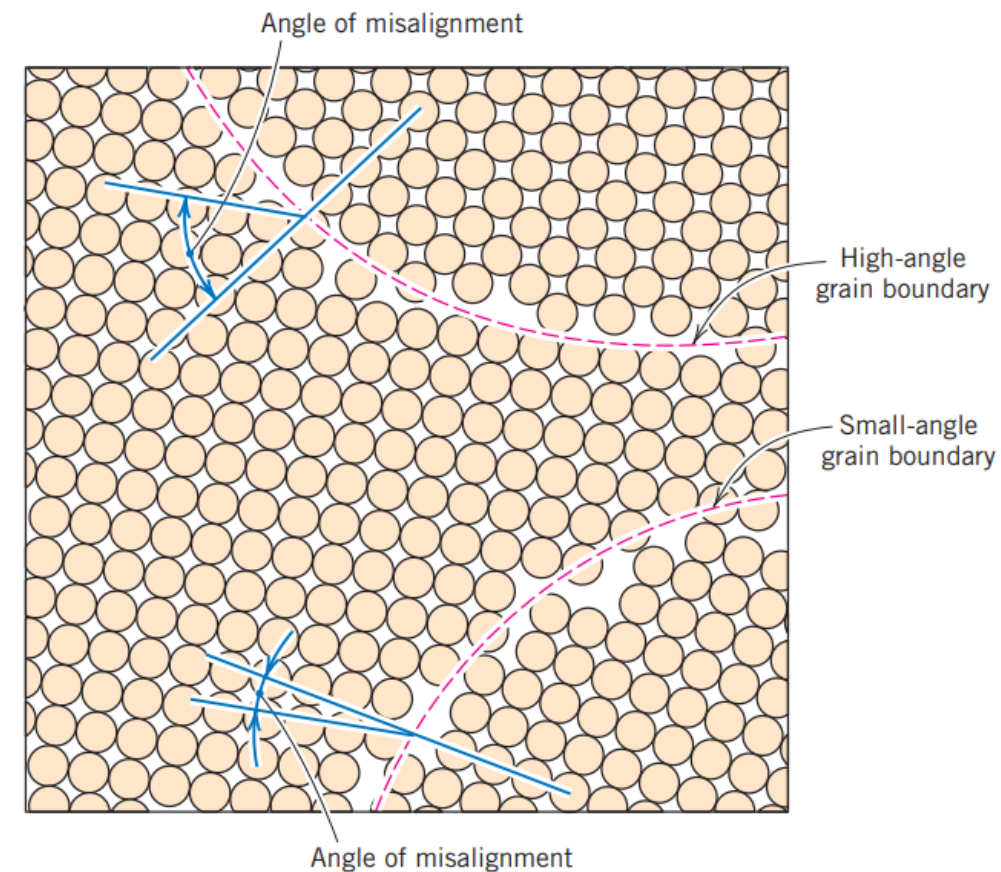
- Grain boundaries;
- Phase boundaries;
- Twin boundaries;
- Stacking faults.

Other types of imperfections in solids

Interfacial Defects

can be same phase

Grain boundaries: regions separating two neighbouring grains having different crystallographic orientations in polycrystals.



- The degree of misalignment in adjacent grains allows distinguishing **small-angle and high-angle grain boundaries**.
- The atoms are bonded less regularly along a grain boundary and consequently there is a high interfacial or grain boundary energy, which is larger for high-angle boundaries.
- Grain boundaries are more chemically reactive than the grains themselves as a consequence of this boundary energy.
- Furthermore, impurity atoms often preferentially segregate along these boundaries because of their higher energy state.
- The total interfacial energy is lower in large or coarse-grained materials than in fine-grained ones because there is less total boundary area in the former.

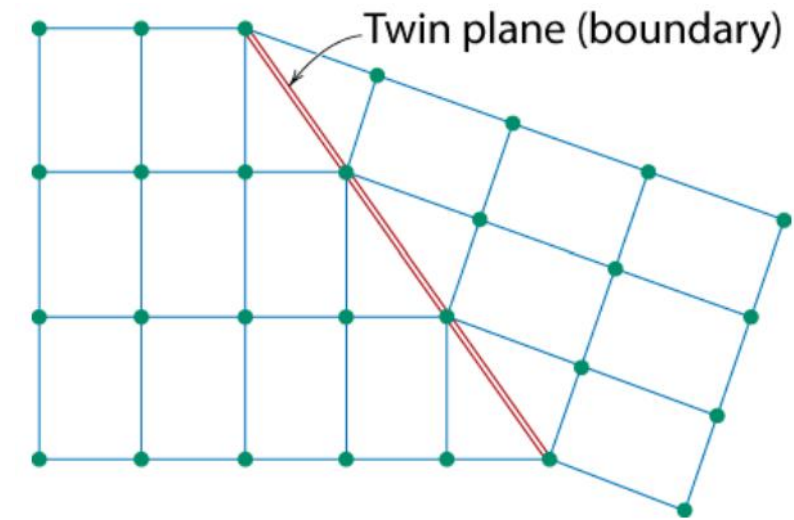
Other types of imperfections in solids

Interfacial Defects

Phase boundaries: regions separating ^{一定是2个不同的相} two different phases, with distinctive physical and chemical characteristics.

Twin boundaries: interface across which there is a specific mirror lattice symmetry. Atoms on one side of the boundary are located in mirror-image positions to those of the atoms on the other side. The two regions are called **twins**.

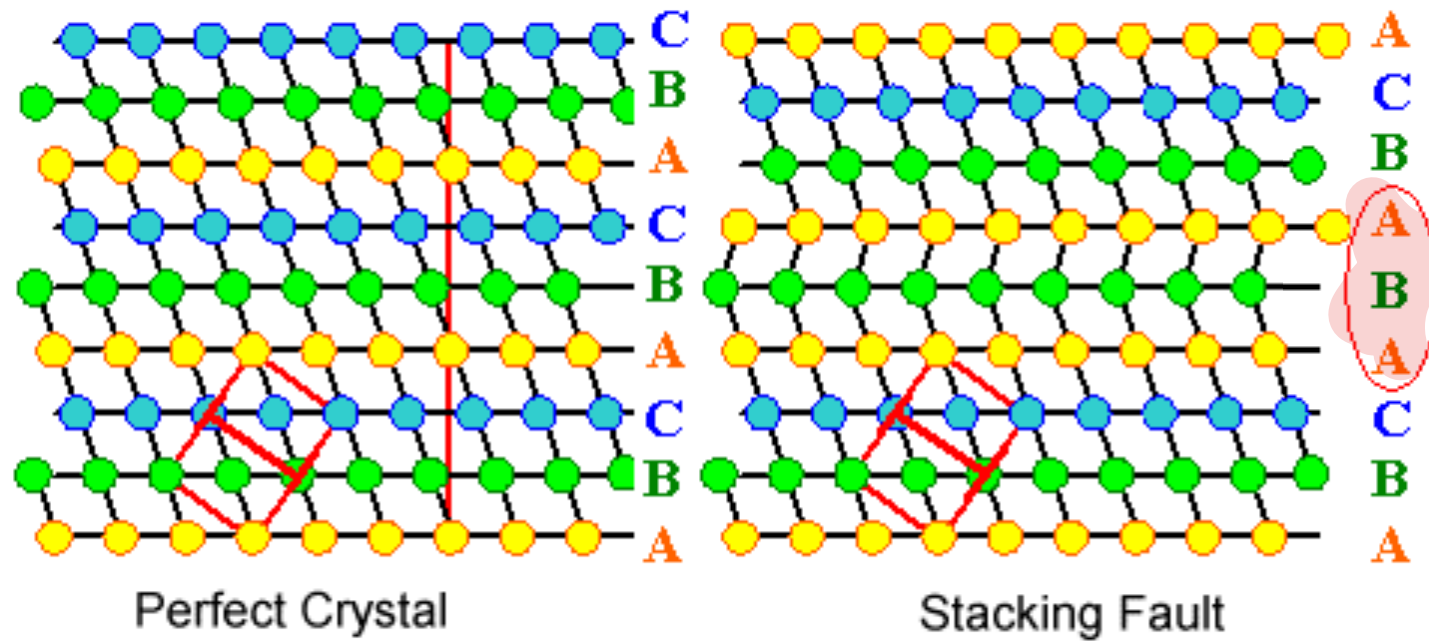
- Twins result from atomic displacements that are produced from applied mechanical shear forces (**mechanical twins**) and also during annealing heat treatments following deformation (**annealing twins**).
- Twinning occurs on a definite crystallographic plane and in a specific direction, both of which depend on the crystal structure.



Other types of imperfections in solids

Interfacial Defects

Stacking faults: crystallographic defects in which the proper order of stacking planes is interrupted.

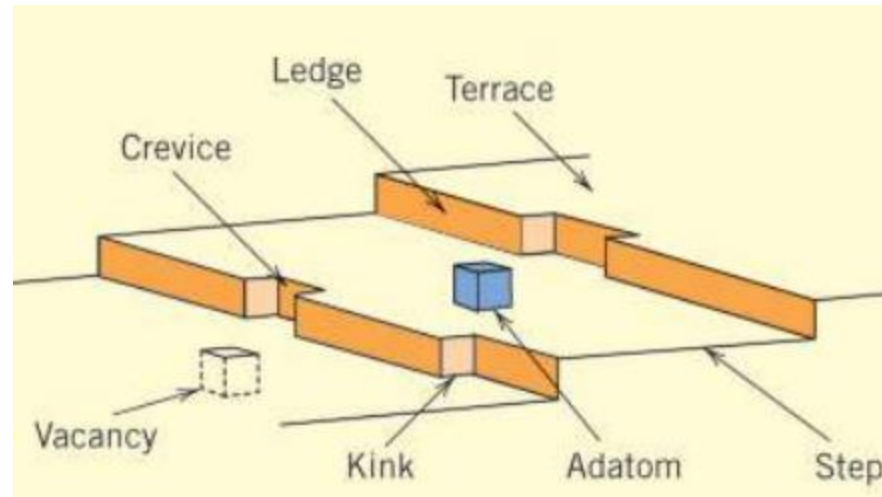


Impact of defects

The presence of defect sometimes is helpful in specific applications, for example in catalysts.

Catalysts: substance that accelerates the rate of a chemical reaction. Molecules taking part to a chemical reaction in a gas or liquid phase can be adsorbed on the catalyst's surface, promoting reactions between the reactants.

Adsorption sites are normally represented by surface defects, such as ledges, kinks, terraces, vacancies, and individual adatoms (i.e., atoms adsorbed on the surface).

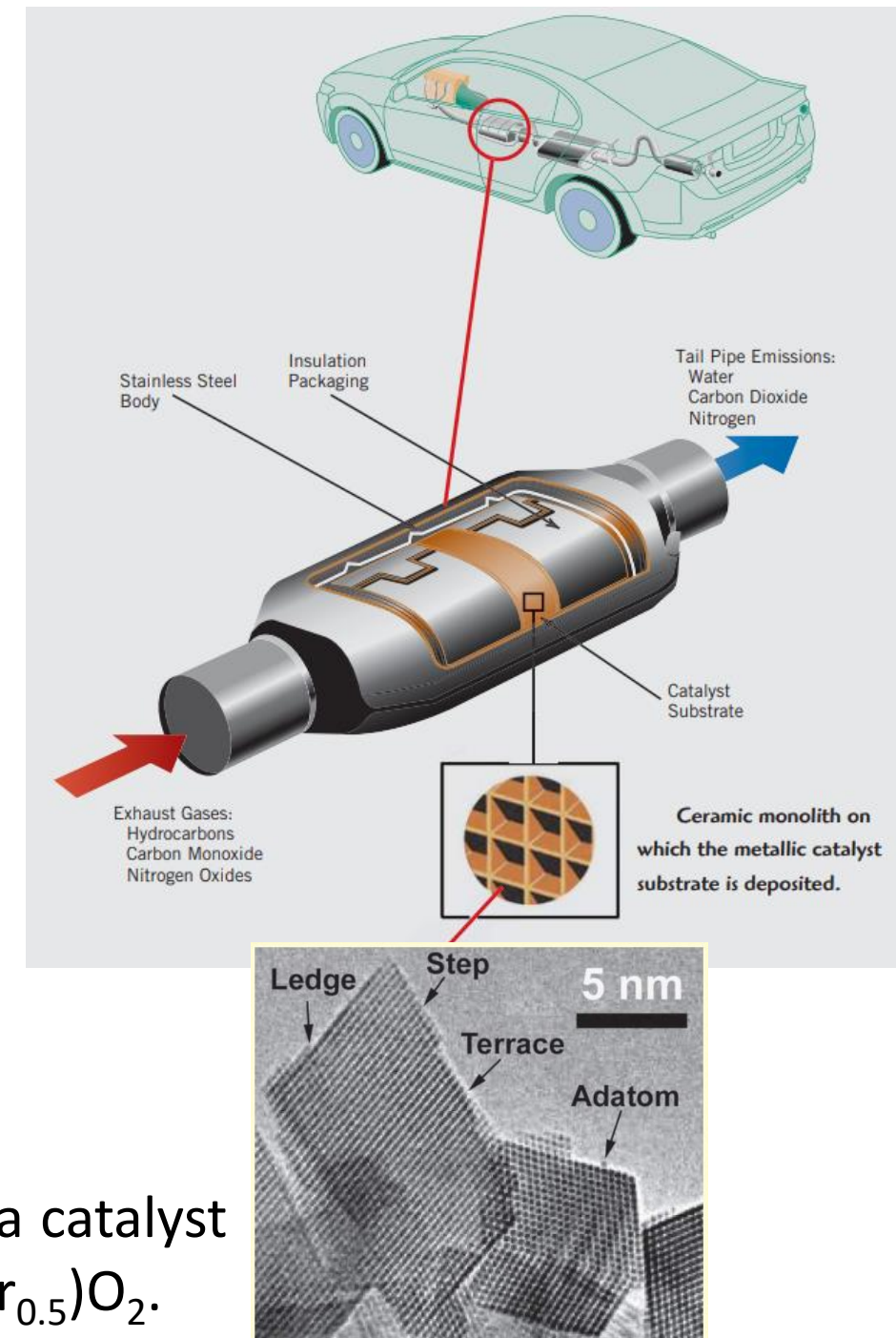


Impact of defects

Catalytic converters in cars: adsorb mixtures of toxic exhaust gases (i.e. CO, and NO_x) and O₂ on their surface defects, promoting the following reactions:

- The NO_x dissociates into N and O atoms;
- The O₂ dissociates into its atomic species;
- Pairs of nitrogen atoms combine to form N₂ molecules, and CO is oxidized to form carbon dioxide (CO₂).
- Unburned hydrocarbons are also oxidized to CO₂ and H₂O.

One of the materials used as a catalyst in these applications is (Ce_{0.5}Zr_{0.5})O₂.



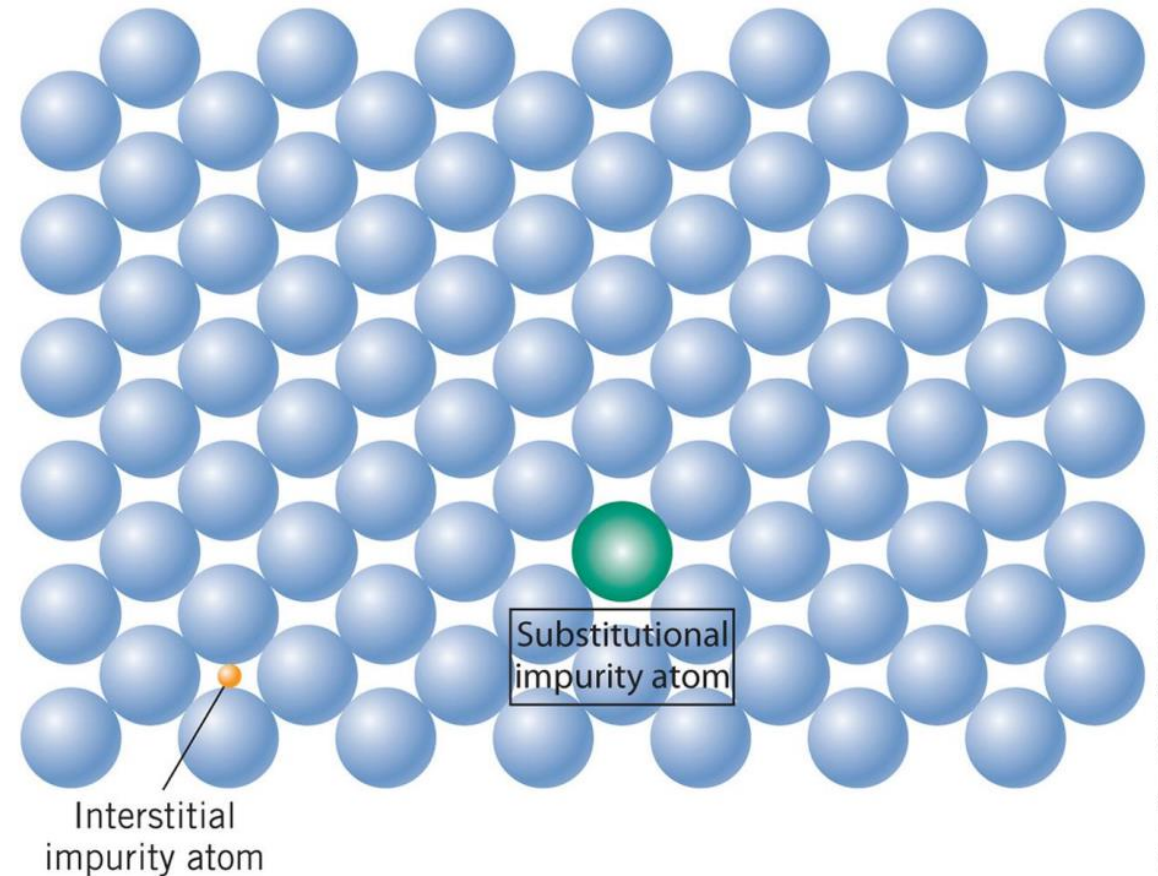
Summary

- Point, Line, and Area defects exist in solids.
- The number and type of defects can be varied and controlled (e.g., temperature controls vacancy concentration).
- Defects affect material properties (e.g., grain boundaries control crystal slip).
- Defects may be desirable or undesirable (e.g. dislocations may be useful or deleterious, depending on whether plastic deformation is desirable or not).

Materials Science 1 – Solid solutions and alloys

Solid solutions

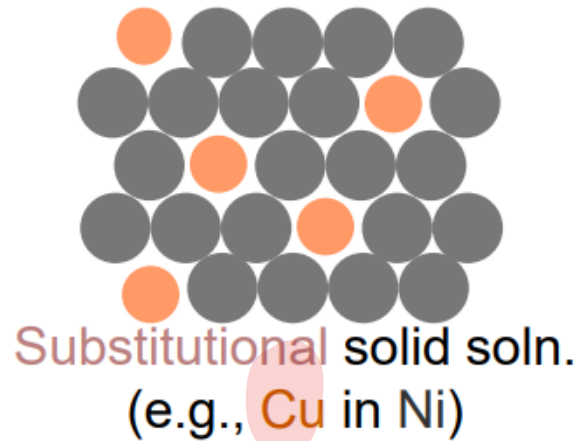
- A solid solution forms when the solute atoms are added to the host material without the formation of secondary phases (i.e. the type of crystal structure is maintained and no new structures are formed).
- A solid solution is also compositionally homogeneous; the impurity atoms are randomly and uniformly dispersed within the solid.
- The solute atoms can locate in different sites of the crystal structure, namely in **vacant sites** and in **interstitial spaces**, depending also on the atomic radius.



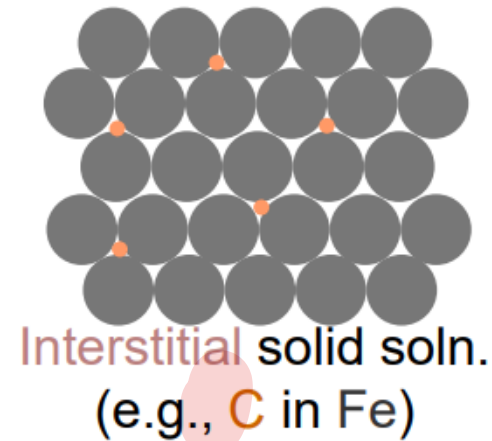
Solid solutions

When a solute atom (B) is added to a hosting solid (A), the following situations can occur:

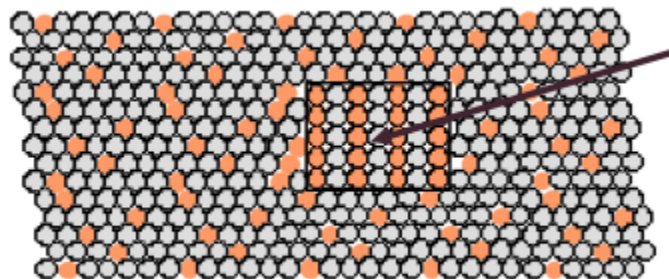
- Solid solution of B in A, with B occupying vacant sites or interstitials:



OR



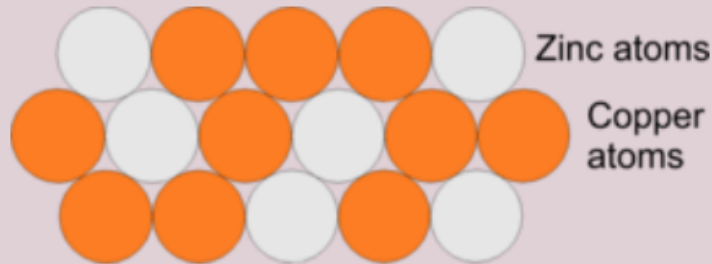
- Solid solution of B in A plus particles of a new phase (second phase). This usually happens for a large amounts of B, exceeding the **solubility limit** of B in A:



Second phase particle
-- different composition
-- often different structure.

Solid solutions

SOLID SOLUTIONS



This Photo by Unknown Author is licensed under [CC BY-SA-NC](#)

- Solid solutions are made of a host (the solvent or matrix) which dissolves the minor component (solute). The ability to dissolve is called solubility
- Solvent: in an alloy, the element or compound present in greater amount
- Solute: in an alloy, the element or compound present in lesser amount
- Solid Solution:
 - Homogeneous
 - maintain crystal structure
 - contain randomly dispersed impurities (substitutional or interstitial)
- Second Phase: as solute atoms are added, new compounds / structures are formed, or solute forms local precipitates
- Whether the addition of impurities results in formation of solid solution or second phase depends the nature of the impurities, their concentration and temperature, pressure

Solid solutions

FACTORS THAT AFFECT THE INCLUSION OF IMPURITIES IN METALS

Several features of the solute and solvent atoms determine the extent of solubility. These are expressed as four **Hume–Rothery rules**:

- **Atomic size factor**: Appreciable solubility occurs when the difference in atomic radii between the two atom types is less than about $\pm 15\%$. Otherwise, the solute atoms create substantial lattice distortions and a new phase forms.
- **Crystal structure**: For appreciable solid solubility the crystal structures for metals of both atom types must be the same.
- **Electronegativity**: The more electropositive one element and the more electronegative the other, the greater is the likelihood that they will form an intermetallic compound instead of a substitutional solid solution.
- **Valences**: Other factors being equal, a metal will have more of a tendency to dissolve another metal of higher valency than one of a lower valency.

Poll – Answers in WeChat

Q2 - What is an alloy?

- a) An alloy is a metal composed of a mixture of elements
- b) An alloy is a metal made of a single elements
- c) An alloy is a metal and polymer combined to make a compound

Poll – Answer

Q2 - What is an alloy?

- a) An alloy is a metal composed of a mixture of elements. Most of alloys are composed of a base metal with small amounts of additives or alloying elements.

Examples of alloys are: steel (Fe-based alloys), bronze (Cu-based alloys) and nickel-based alloys.

Solid solutions

Example 5: Application of Hume-Rothery rules – Solid Solutions

According to the information in the table below, would you predict more Al or Ag to dissolve in Zn?

<i>Element</i>	<i>Atomic Radius (nm)</i>	<i>Crystal Structure</i>	<i>Electro- nega- tivity</i>	<i>Valence</i>
Cu	0.1278	FCC	1.9	+2
C	0.071			
H	0.046			
O	0.060			
Ag	0.1445	FCC	1.9	+1
Al	0.1431	FCC	1.5	+3
Co	0.1253	HCP	1.8	+2
Cr	0.1249	BCC	1.6	+3
Fe	0.1241	BCC	1.8	+2
Ni	0.1246	FCC	1.8	+2
Pd	0.1376	FCC	2.2	+2
Zn	0.1332	HCP	1.6	+2

Solid solutions

Example 5: Application of Hume-Rothery rules – Solid Solutions - Solution

According to the information in the table below, would you predict more Al or Ag to dissolve in Zn?

<i>Element</i>	<i>Atomic Radius (nm)</i>	<i>Crystal Structure</i>	<i>Electro- nega- tivity</i>	<i>Valence</i>
Cu	0.1278	FCC	1.9	+2
C	0.071			
H	0.046			
O	0.060			
Ag	0.1445	FCC	1.9	+1
Al	0.1431	FCC	1.5	+3
Co	0.1253	HCP	1.8	+2
Cr	0.1249	BCC	1.6	+3
Fe	0.1241	BCC	1.8	+2
Ni	0.1246	FCC	1.8	+2
Pd	0.1376	FCC	2.2	+2
Zn	0.1332	HCP	1.6	+2

Al would dissolve in Zn more than Ag, because it has more similar atomic radius, more similar electronegativity and closer valence to Zn.

The biggest limit is the different crystal structure and higher valence than Zn.

Solid solutions

Example 6: Application of Hume-Rothery rules – Solid Solutions

Discuss with your classmates which of the following systems (i.e., pair of metals) would you expect to exhibit complete solid solubility? Justify your answers.

- (a) Cr-V
- (b) Mg-Zn
- (c) Zn in Al
- (d) Zn in Cu

Requirements: The four Hume-Rothery rules must be satisfied:

- 1) the difference in atomic radii between elements (DR%) must be less than $\pm 15\%$
- 2) the crystal structures must be the same
- 3) the electronegativities must be similar
- 4) the valences should be the same

Solid solutions

Example 6: Application of Hume-Rothery rules – Solid Solutions - Solution

Discuss with your classmates which of the following systems (i.e., pair of metals) would you expect to exhibit complete solid solubility? Justify your answers.

- (a)Cr-V
- (b)Mg-Zn
- (c)Zn in Al
- (d)Zn in Cu

Metal	Atomic Radius (nm)	Crystal Structure	Electronegativity	Valence
Cr	0.125	BCC	1.6	+3
V	0.132	BCC	1.5	+3, +5

Chromium and vanadium are completely soluble in one another $\left| \frac{r_{Cr} - r_V}{r_{Cr}} \right| < 0.15$

Metal	Atomic Radius (nm)	Crystal Structure	Electronegativity	Valence
Mg	0.160	HCP	1.3	+2
Zn	0.133	HCP	1.7	+2

Magnesium and zinc are not completely soluble in one another, primarily because of the difference in atomic radii. $\left| \frac{r_{Mg} - r_{Zn}}{r_{Mg}} \right| > 0.15$

Materials Science 1 – Specification of composition

Compositions and concentrations in alloys

It is often necessary to express the composition of an alloy in terms of the concentration, or fraction of its constituents.

There are two most common ways to specify the composition of an alloy:

- **Weight (or mass) percentage (wt%);**
- **Atom percentage.**

Example:

Consider an alloy containing two hypothetical atoms denoted by 1 and 2. The concentration of element 1 the alloy is given by:

weight percent

$$C_1 = \frac{m_1}{m_1 + m_2} \times 100$$

m_1 : mass of component 1;
 m_2 : mass of component 2.

atom percent

$$C'_1 = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100$$

n_{m1} : number of moles of component 1;
 n_{m2} : number of moles of component 2;
 A_i : atomic weight of element i.

$$n_{mi} = \frac{m_i}{A_i}$$

Compositions and concentrations in alloys

Composition conversions

To obtain atomic percentage C_1' from weight percentage C_1 , and C_1 from C_1' , one can use the following relationships:

$$C_1' = \frac{C_1 A_2}{C_1 A_2 + C_2 A_1} \times 100$$

Conversion of weight percent to
atom percent

$$C_2' = \frac{C_2 A_1}{C_1 A_2 + C_2 A_1} \times 100$$

$$C_1 = \frac{C_1' A_1}{C_1' A_1 + C_2' A_2} \times 100$$

Conversion of atom percent to
weight percent

$$C_2 = \frac{C_2' A_2}{C_1' A_1 + C_2' A_2} \times 100$$

Compositions and concentrations in alloys

Average density of an alloy

$$\rho_{\text{ave}} = \frac{100}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}}$$

Derivation

$$\rho_{ave} = \frac{m_{tot}}{V_{tot}} = \frac{m_1 + m_2}{V_1 + V_2} = \frac{m_{tot}}{\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}} = \frac{m_{tot}}{\frac{m_1\rho_2 + m_2\rho_1}{\rho_1\rho_2}} = \frac{m_{tot}}{\frac{C_1 m_{tot}\rho_2 + C_2 m_{tot}\rho_1}{\rho_1\rho_2}} = \frac{\rho_1\rho_2}{C_1\rho_2 + C_2\rho_1} = \frac{1}{\frac{C_1\rho_2 + C_2\rho_1}{\rho_1\rho_2}} = \frac{1}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}}$$

In terms of volume fractions F_1 and F_2

$$\rho_{ave} = \frac{m_{tot}}{V_{tot}} = \frac{m_1 + m_2}{V_1 + V_2} = \frac{\rho_1 V_1 + \rho_2 V_2}{V_1 + V_2} = \rho_1 F_1 + \rho_2 F_2$$

Compositions and concentrations in alloys

Average atomic weight in an alloy

$$A_{\text{ave}} = \frac{100}{\frac{C_1}{A_1} + \frac{C_2}{A_2}}$$

Derivation

$$A_{\text{ave}} = \frac{m_{\text{tot}}}{n_{\text{tot}}} = \frac{m_1 + m_2}{n_1 + n_2} = \frac{C_1 m_{\text{tot}} + C_2 m_{\text{tot}}}{\frac{m_1}{A_1} + \frac{m_2}{A_2}} = \frac{m_{\text{tot}}(C_1 + C_2)}{\frac{m_1}{A_1} + \frac{m_2}{A_2}} = \frac{100}{\frac{m_1}{m_{\text{tot}} A_1} + \frac{m_2}{m_{\text{tot}} A_2}} = \frac{100}{\frac{C_1}{A_1} + \frac{C_2}{A_2}}$$

Compositions and concentrations in alloys

CLASS ACTIVITY

Work with a classmate or on your own

Send your answers via WeChat

Compositions and concentrations in alloys

Example 1:

What is the composition, in atom percent, of an alloy that consists of 92.5 wt% Ag and 7.5 wt% Cu?

Example 2:

What is the composition, in weight percent, of an alloy that consists of 5 at% Cu and 95 at% Pt?

Example 3:

A hypothetical alloy is composed of 25 wt% of metal A and 75 wt% of metal B. If the densities of metals A and B are 6.17 and 8.00 g/cm³, respectively, and their respective atomic weights are 171.3 and 162.0 g/mol, determine whether the crystal structure for this alloy is simple cubic, face-centred cubic, or body-centred cubic. Assume a unit cell edge length of $a=0.332$ nm.

Compositions and concentrations in alloys

Example 1 solution:

What is the composition, in atom percent, of an alloy that consists of 92.5 wt% Ag and 7.5 wt% Cu?

Atomic weights

$$A_{Ag} = 107.87 \text{ g/mol}$$

$$A_{Cu} = 63.55 \text{ g/mol}$$

1. Calculate the composition of Ag

$$C'_{Ag} = \frac{C_{Ag}A_{Cu}}{C_{Ag}A_{Cu} + C_{Cu}A_{Ag}} \times 100 = \frac{(92.5)(63.55 \text{ g/mol})}{(92.5)(63.55 \text{ g/mol}) + (7.5)(107.87 \text{ g/mol})} \times 100$$

$$C_{Ag} = 87.9 \text{ at\%}$$

2. Calculate the composition of Cu

$$C'_{Cu} = \frac{C_{Cu}A_{Ag}}{C_{Ag}A_{Cu} + C_{Cu}A_{Ag}} \times 100 = \frac{(7.5)(107.87 \text{ g/mol})}{(92.5)(63.55 \text{ g/mol}) + (7.5)(107.87 \text{ g/mol})} \times 100$$

$$C_{Cu} = 12.1 \text{ at\%}$$

Compositions and concentrations in alloys

Example 2 solution:

What is the composition, in weight percent, of an alloy that consists of 5 at% Cu and 95 at% Pt?

$$C_{Pt} = \frac{C_{Pt} A_{Pt}}{C_{Pt} A_{Pt} + C_{Cu} A_{Cu}} \times 100 = \frac{185.25}{185.25 + 3.2} \times 100 = 98.3$$
$$C_{Cu} = \frac{C_{Cu} A_{Cu}}{C_{Pt} A_{Pt} + C_{Cu} A_{Cu}} \times 100 = 1.7$$

Compositions and concentrations in alloys

Example 3 solution:

A hypothetical alloy is composed of 25 wt% of metal A and 75 wt% of metal B. If the densities of metals A and B are 6.17 and 8.00 g/cm³, respectively, and their respective atomic weights are 171.3 and 162.0 g/mol, determine whether the crystal structure for this alloy is simple cubic, face-centred cubic, or body-centred cubic. Assume a unit cell edge length of $a=0.332$ nm.

Step 1:

The specific cubic crystal structure can be identified by the number of atoms of A and B contained in the cubic volume a^3 .

From the general formula of the crystal structure's density:

$$\rho = \frac{nA}{V_C N_A}$$

It follows:

$$\rho_{ave} = \frac{nA_{av}}{V_C N_A} = \frac{nA_{av}}{a^3 N_A}$$



$$n = \frac{\rho_{ave} a^3 N_A}{A_{ave}}$$

Compositions and concentrations in alloys

Example 3 solution:

A hypothetical alloy is composed of 25 wt% of metal A and 75 wt% of metal B. If the densities of metals A and B are 6.17 and 8.00 g/cm³, respectively, and their respective atomic weights are 171.3 and 162.0 g/mol, determine whether the crystal structure for this alloy is simple cubic, face-centred cubic, or body-centred cubic. Assume a unit cell edge length of 0.332 nm.

The density and atomic weight can be calculated by:

$$\rho_{\text{ave}} = \frac{100}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}}$$

$$A_{\text{ave}} = \frac{100}{\frac{C_1}{A_1} + \frac{C_2}{A_2}}$$

Substituting these in the previous equation, n can be found:

$$n = \frac{\left(\frac{100}{\frac{C_A}{\rho_A} + \frac{C_B}{\rho_B}} \right) a^3 N_A}{\left(\frac{100}{\frac{C_A}{A_A} + \frac{C_B}{A_B}} \right)}$$

Compositions and concentrations in alloys

Example 3 solution:

A hypothetical alloy is composed of 25 wt% of metal A and 75 wt% of metal B. If the densities of metals A and B are 6.17 and 8.00 g/cm³, respectively, and their respective atomic weights are 171.3 and 162.0 g/mol, determine whether the crystal structure for this alloy is simple cubic, face-centred cubic, or body-centred cubic. Assume a unit cell edge length of a=0.332 nm.

$$n = \frac{\left(\frac{100}{\frac{25 \text{ wt}\%}{6.17 \text{ g/cm}^3} + \frac{75 \text{ wt}\%}{8.00 \text{ g/cm}^3}} \right) (3.32 \times 10^{-8} \text{ cm})^3 (6.022 \times 10^{23} \text{ atoms/mol})}{\left(\frac{100}{\frac{25 \text{ wt}\%}{171.3 \text{ g/mol}} + \frac{75 \text{ wt}\%}{162.0 \text{ g/mol}}} \right)}$$
$$n = 1.00 \text{ atom/unit cell}$$

Therefore, on the basis of this value, the crystal structure is *simple cubic*.